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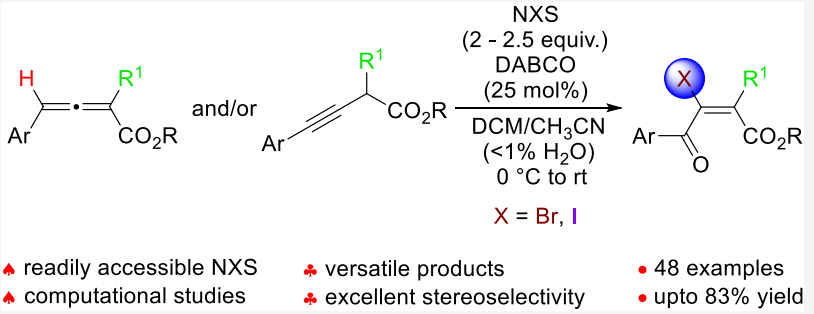
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Electrophilic Halogenation of Allenoates and 3-Alkynoates: Synthesis of 1,4-Dicarbonyl (*E*)-3-Haloalkenes and Mechanistic Investigations

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ABSTRACT: Allenoates traditionally provide halobutenolides upon reaction with halonium ions via electrophile-assisted halolactonization. Herein, an unusual electrophilic halogenation of di- and trisubstituted allenoates and 3-alkynoates is demonstrated with *N*-halosuccinimides under DABCO promotion. The protocol affords densely functionalized 1,4-dicarbonyl 3-haloalkenes in good yields with excellent (*E*)-stereoselectivity (up to 83% yield, >20:1 dr). The allenoates are presumed to form γ-haloallenoate intermediates, which further react with halonium ions and provide the desired scaffolds. The role of the nucleophilic base in the unusual transformation is demystified through control experiments and computational studies. The deliverables are identified as good synthons for various synthetic group transformations and valuable targets for biologically active pyridazine scaffolds.

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INTRODUCTION

Allenoates are versatile synthons in organic synthesis, offering a wide range of possibilities for numerous synthetic transformations.[1] Their unique electronic and structural properties make them fascinating molecules with multifaceted reactivity, often considered as an amalgam of alkene and alkyne.[2] The allenoate scaffolds are renowned for a variety of addition reactions, annulations, and rearrangement reactions in organic synthesis.[3] In addition, they have also received significant attention for halonium-assisted lactonization, which produces 4-halo-2(5H)-furanone (Scheme 1a).[4−7] This approach was well studied with electrophilic iodinations,[4] brominations,[5] and fluorinations.[6] Interestingly, Ma and co-workers observed the formation of 3-fluoro-4-oxo-2(*E*)-alkenoates during the synthesis of fluorobutenolides with Selectfluor.[6] However, a protocol that could provide analogous bromo- or iodoenoates is synthetically more useful and highly demanded. If successful, the haloalkene and 1,4-dicarbonyl components in the resultant scaffolds could efficiently be used for target-oriented metal-catalyzed cross-coupling reactions and heterocycle synthesis.[8] On the other hand, 3-alkynoates are also employed in iodolactonization that delivers 4-iodo-2(3H)-furanone regioisomer.[7] In this context, we wondered whether we could obtain a convergent product from a mixture of allenoates and 3-alkynoates via electrophilic halogenation. Indeed, this approach is practically more effective as the significant allenoate synthetic protocols tend to give 3-alkynoates in variable amounts along with the desired allenoates.[9] We envisioned that a base-mediated in situ isomerization of 3-alkynoates to allenoates could address this issue provided allenoate reactivity toward an electrophile is better than that of alkynoates.

Recently, we demonstrated the applicability of alkynyl hydrazones for the synthesis of disubstituted allenoates via a base-promoted Wolff−Kishner reduction, followed by isomerization.[10] Hydrazones are also potential to provide dihaloallenoates upon reaction with two equivalents of *N*-halosuccinimides. Gratifyingly, the resultant dihaloallenoates smoothly react with another equivalent of halonium source and afford a divergent dihaloenoate core.[11] However, harnessing stereo-

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Scheme 1. Background of the Research

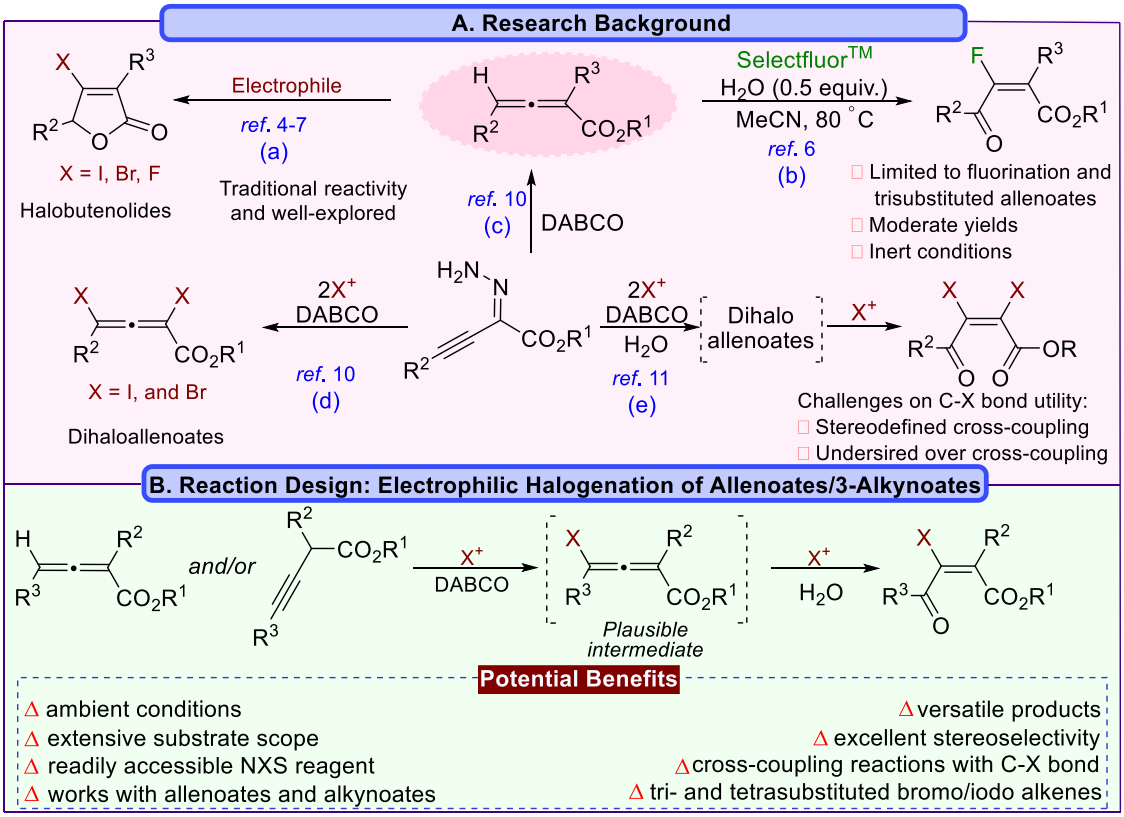
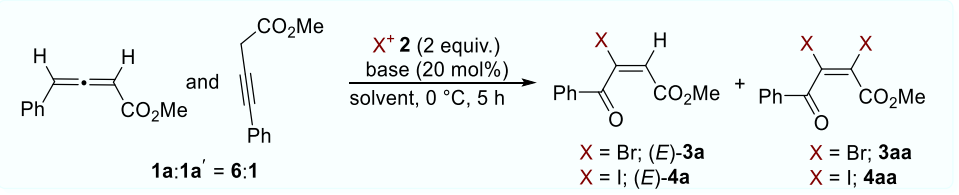


Table 1. Optimization Studies[a]



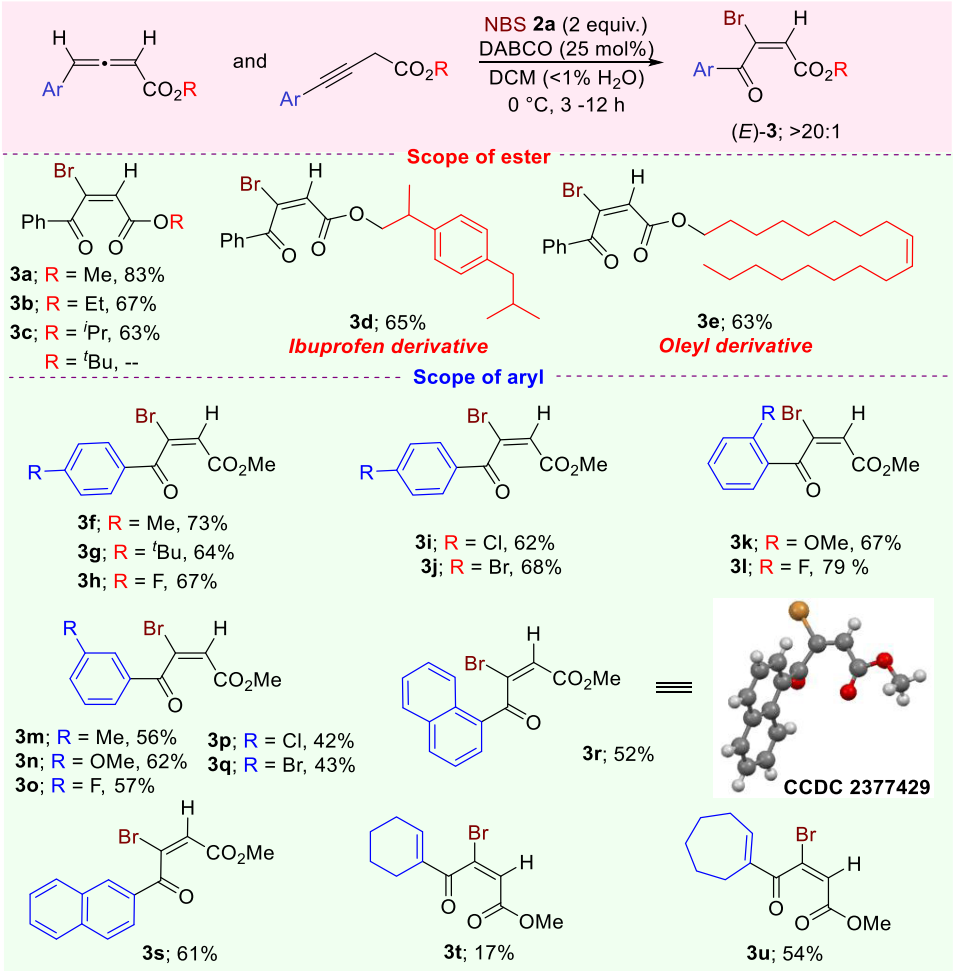
s. no.[a]	2	base	solvent	conv. (%)[b]	(E:Z)-3a	3a/4a (%)[b]	3aa/4aa (%)[b]
1	**2a**	DMAP	DCM	81	>20:1	64	3
2	**2a**	DABCO	DCM	>99	>20:1	78	2
3	**2a**	K_2CO_3	DCM	>92	>20:1	39	6
4	**2a**	DABCO	MeCN	90	16:1	67	1
5[c]	**2a**	DABCO	DCM	>99	5:1	77[d]	12
6[e]	**2a**	DABCO	DCM	>99	>20:1	66[d]	14
7[f]	**2a**	DABCO	DCM	>99	>20:1	83[d]	3
8[f]	**2b**	DABCO	DCM	>99	>20:1	78[d]	3
9[f]	**2c**	DABCO	DCM	>99	8:1	61[d]	3
10[f]	**2d**	DABCO	DCM	>99	>20:1	34[d]	17
11[f]	**2e**	DABCO	DCM	>99	>20:1	37	6
12[f]	**2e**	DABCO	DCM	53	5:1	62[d]	3
13[f,g]	**2e**	DABCO	MeCN	78	>20:1	75	2

[a]Reaction conditions: **1a**:**1a**′ (0.1 mmol), **2a** (2 equiv), solvent (0.05 M), and base (20 mol %) at 0 °C. [b]Determined by crude ^{1}H NMR analysis using $CHPh_3$ as an internal standard. [c]0 to 40 °C. [d]Isolated yield. [e]2.4 equiv of **2a**. [f]25 mol % base. [g]At 25 °C with **2e** (2.5 equiv). **2a** = *N*-bromosuccinimide; **2b** = *N*-bromophthalimide; **2c** = 1,3-dibromo-5,5-dimethylimidazolidine-2,4-dione; **2d** = 2,4,4,6-tetrabromo-2,5-cyclohexadienone; **2e** = *N*-iodosuccinimide.

defined sequential cross-coupling reactions on dihaloenoates is a formidable challenge since the reactions could provide a mixture of inseparable regioisomers. Given the difficulties, we wondered whether we could synthesize a monohaloenoate core as a better coupling template since the competitive regio coupling or undesired over-cross-coupling issues will not appear with these motifs. In this context, we studied an unusual electrophilic halogenation of allenoates and 3-alkynoates with NBS/NIS reagents under DABCO promotion to provide highly valued 1,4-dicarbonyl bromo and iodoalkenes (Scheme

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Scheme 2. Electrophilic Bromination of Disubstituted Allenoates and 3-Alkynoates[a,b]



[a]Reaction conditions: **1a**:**1a**′ (1 equiv), **2a** (2 equiv), DCM (0.05 M), and DABCO (25 mol %) at 0 °C. [b]Isolated yield.

1b). The protocol is presumed to proceed through haloallenoate intermediates,[12] and the reaction mechanism is elucidated through control experiments and computational studies.

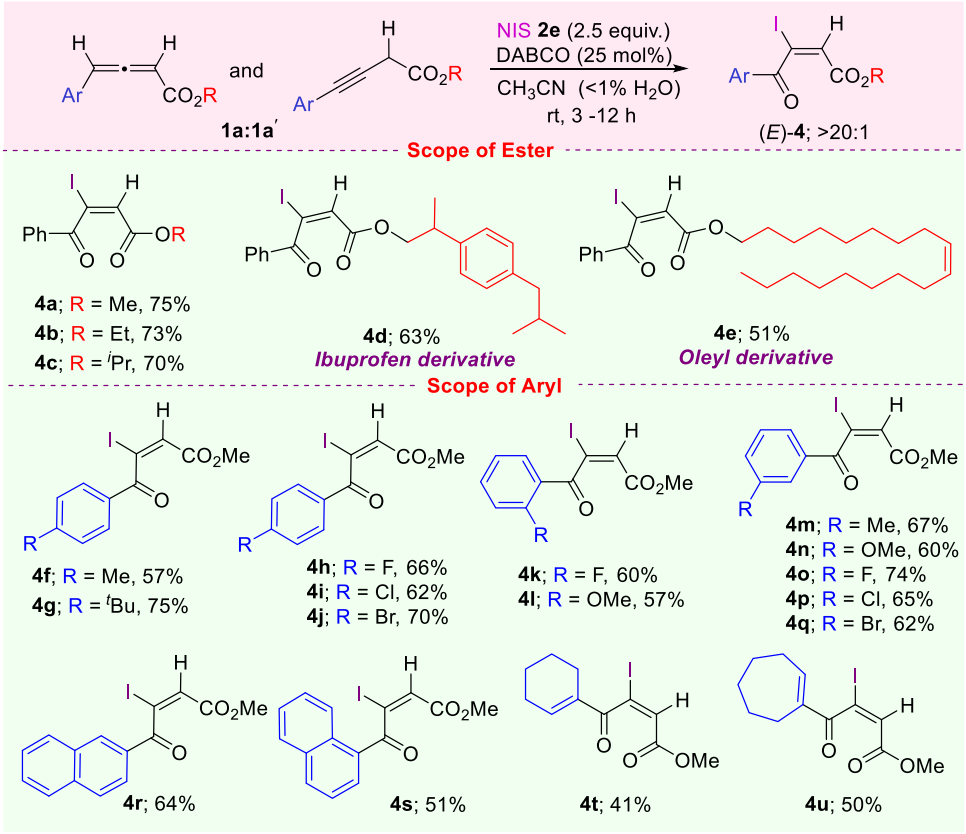
RESULTS AND DISCUSSION

First, we screened the various organic and inorganic bases and found that 1,4-diazabicyclo[2.2.2]octane (DABCO) was the most efficient base which provided (*E*)-**3a** in 78% yield with excellent selectivity (>20:1, Table 1, entries 1−3).[13] The reactions have also provided traces of dihaloenoate **3aa**. Further optimization of the solvents did not improve the yield of **3a**, while a significant drop in stereoselectivity (5:1) was observed when the temperature was raised to 40 °C (entry 4−5). Next, we attempted to improve the yield by increasing the *N*-bromosuccinimide (NBS) reagent loading, but the reaction afforded (*E*)-**3a** in a decreased yield (66%, entry 6). Delightfully, a slight increase in the DABCO loading (25 mol %) improved the yield of the reaction and **3a** was isolated in 83% yield (entry 7). Next, we examined the reaction outcome with the other electrophilic bromine sources (**2b**−**2d**) and found that the most common NBS reagent outperformed the others (entries 7, 8−10). Further, we attempted to validate the optimal conditions of electrophilic bromination to analogous iodination with the *N*-iodosuccinimide (NIS) reagent; however, we observed a drastic decrease in the yield of **4a** (37%, entry 11). It encouraged us to further fine-tune the optimized conditions by screening solvents, temperatures, and the NIS loading.[13] To our delight, we found a good yield of **4a** in CH_3CN with 2.5 equiv of the NIS reagent (75% yield, entry 13).

With the optimal reaction conditions, we investigated the substrate scope with a series of allenoates and the NBS/NIS reagents (Scheme 2). First, we explored the reactions by varying the alkyl group (Et and ^{i}Pr) on the ester component of the allenoates. To our delight, we observed the corresponding products **3b**−**3c** in good yields (63−67%). While the ester containing a sterically crowded *tert*-butyl group did not provide the desired product.[14] However, the allenoates tethered to biologically active ibuprofen and naturally occurring oleyl alcohol scaffolds provided the desired products (**3d**−**3e**) in good yields (63−65%). Then, we turned our attention to screen allenoates with different aryl groups. We varied the electron-donating and withdrawing groups at the *p*-position of the aryl group, and the reactions yielded the corresponding bromoenoates **3f**−**3j** in 62−73% yields. The aryl groups also tolerated the steric substitution at the *ortho* position, and the corresponding *o*-substituted alkenyl carboxylates **3k**−**3l** were obtained in good yields (67−79%). In addition, allenoates containing *m*-substituted aryls reacted smoothly and the

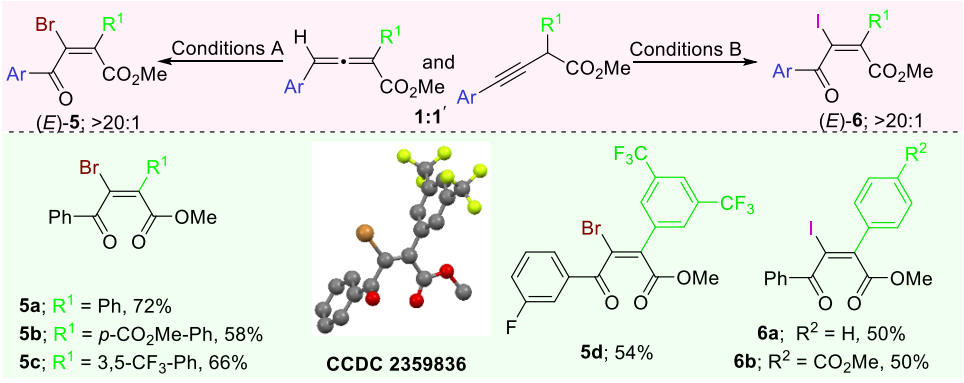
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Scheme 3. Electrophilic Iodination of Disubstituted Allenoates and 3-Alkynoates[a,b]



[a]Reaction conditions: **1a**:**1a**′ (1 equiv), **2e** (2.5 equiv), CH_3CN (0.05 M), and DABCO (25 mol %) at 25 °C. [b]Isolated yield.

Scheme 4. Electrophilic Bromination and Iodination of Trisubstituted Allenoates[a−c]



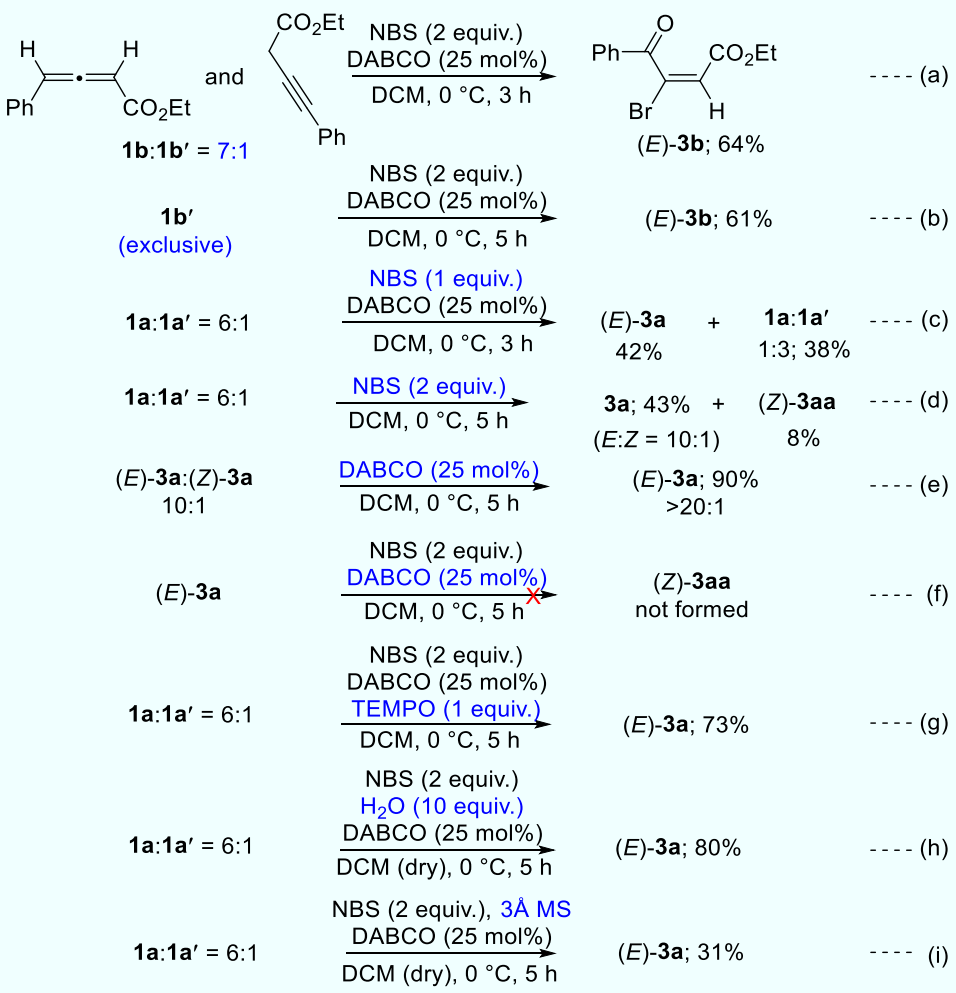
[b]Isolated yield. [a]**Conditions A**: **1a**:**1a**′ (1 equiv), **2a** (2 equiv), DCM (0.05 M), and DABCO (25 mol %) at 0 °C. [c]**Conditions B**: **1a**:**1a**′ (1 equiv), **2e** (2.5 equiv), CH_3CN (0.05 M), and DABCO (25 mol %) at 25 °C.

corresponding products **3m**−**3q** were isolated in 42−62% yields. Notably, we could also incorporate the *α*- and *β*-naphthyl groups into the products, and the reactions resulted in **3r** and **3s** in 52 and 61% yields, respectively. Next, we screened the cyclohexenyl and cycloheptenyl-substituted allenoates under optimized conditions. Remarkably, the impact of ring size on chemical yields was significant as we observed a poor yield for cyclohexenyl product **3t**,[15] whereas cycloheptenyl product **3u** was isolated in 54% yield. The stereochemistry of **3r** was unequivocally established from X-ray structural analysis, and their analogs were assigned accordingly.

Next, considering the high demand for iodoalkenes, we executed an analogous substrate scope with the NIS reagent under the optimized conditions (Scheme 3). We obtained the Et and ^{i}Pr substituted iodoenoates **4b**−**4c** in comparable yields (70−73%). Next, we screened the allenoates derived from ibuprofen and oleyl alcohol, which resulted in **4d**−**4e** with 51−63% yields. The allenoates that contain different substitutions at *p*-, *o*-, and *m*-substituted aryls were also screened, and the reactions yielded **4f**−**4q** in 57−75% yields with excellent selectivity. Additionally, 1-naphthyl and 2-naphthyl allenoates were compatible with standard reaction conditions and provided the respective **4r**−**4s** products with 51−64% yields.

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Scheme 5. Control Experiments



The optimized conditions are also compatible with the cyclohexenyl and cycloheptenyl-substituted allenoates and provides the desired iodoenoates **4t**−**4u**.

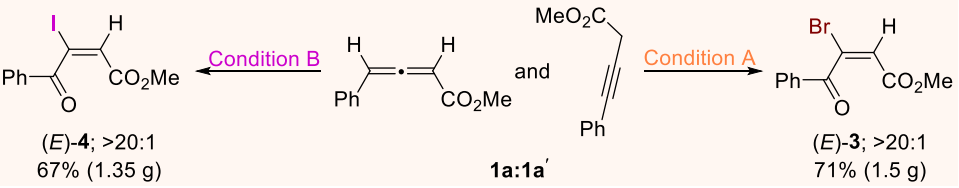
To broaden the substrate scope of electrophilic halogenation, we intend to investigate the reactivity of trisubstituted allenoates (Scheme 4). The substrates are synthesized through metal-free cross-coupling of alkynyl diazoacetates and boronic acids under photochemical conditions.[13,16] Delightfully, we successfully obtained the desired products, and the corresponding tetra-substituted bromo or iodoenoates **5**/**6** were obtained in 50−72% yields. The stereochemistry of **5c** was unambiguously established from X-ray structural analysis.

We attempted a few control experiments to gain insight into the reaction mechanism (Scheme 5). First, we intend to study the impact of the isomeric purity of the allenoates on the chemical yield of the reactions. Hence, we synthesized alkynoate and allenoate in major quantities and executed the reactions under optimized conditions.[10,17] Delightfully, we obtained the desired product in comparable yields, despite the variations in quantities of allenoate and alkynoate (eq a, b). This shows that yields of the desired product are independent of the isomeric purity of the allenoates. Next, we executed the optimized conditions with one equivalent of the NBS reagent which produced **3a** only in 42% yield (eq c). It clearly emphasizes the necessity of an additional equivalent of NBS to obtain the quantitative amount of **3a**. It is also observed that the reaction transpired even in the absence of a catalyst. However, the desired product **3a** was merely observed in moderate yield and selectivity (43%; 10:1 dr), and the side product **3aa** was noticed in 8% yield (eq d). Further, we investigated a probable Lewis base-promoted isomerization of (*E*)-**3a** to (*Z*)-**3a** under standard conditions. The reaction shows an in situ isomerization (eq e).[18] Next, we investigated a sequential bromination of monobromoenoate **3a** to dibromoenoate **3aa** under optimized conditions. However, the reaction did not show any in situ transformation (eq f). Probably, the side product **3aa** could have formed through electrophilic bromination of dibromoallenoate which might have formed in traces under the investigated conditions.[11] Next, we performed a reaction with TEMPO to identify any probable radical species in the reactions (eq g). The reaction provided **3a** in a comparable yield which ruled out the radical pathway. Finally, we carried out a reaction in a dry solvent with an addition of H_2O (10 equiv), which yielded **3a** with an 80% yield (eq h). Whereas a significant decrease in yield was noticed when we performed the reaction in the presence of molecular sieves (eq i).

Furthermore, we performed gram-scale reactions with NBS and NIS reagents to demonstrate the scalability and robustness of the optimized reaction conditions. The reactions provided (*E*)-**3a** and (*E*)-**4a** in 71 and 67% yields, respectively (Scheme 6).

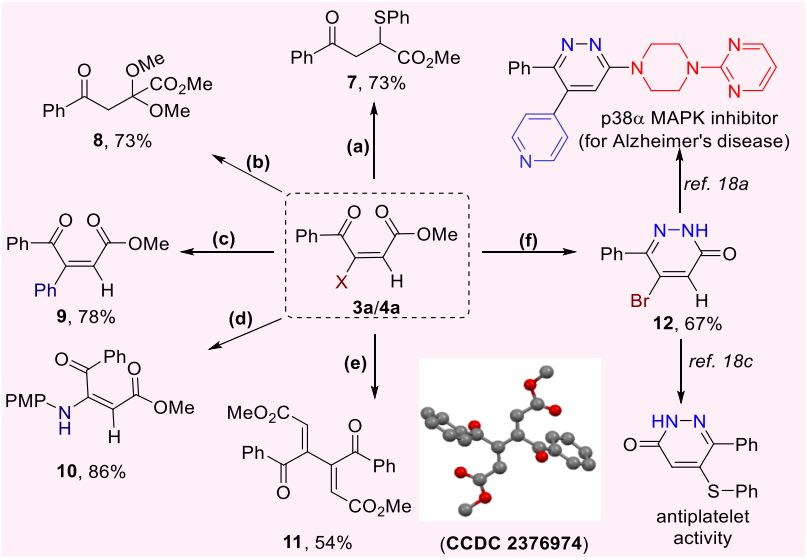
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Scheme 6. Gram-Scale Synthesis of 1,4-Dicarbonyl (*E*)-3-Haloalkenes[a-c]



[b]Isolated yield. [c]**Conditions B**: **1a**:**1a**′ (7.84 mmol, 1 equiv), **2e** (19.6 mmol, 2.5 equiv), CH_3CN (0.05 M), and DABCO (25 mol %) at 25 °C. [a]**Conditions A**: **1a**:**1a**′ (7.84 mmol, 1 equiv), **2a** (15.6 mmol, 2 equiv), DCM (0.05 M), and DABCO (25 mol %) at 0 °C.

Scheme 7. Synthetic Transformations[a]



[a]Reaction conditions: (a) PhSH, K_2CO_3; (b) Cs_2CO_3/MeOH; (c) $PhB(OH)_2$, $PdCl_2(PPh_3)_2$, K_2CO_3; (d) $Pd_2(dba)_3$, Xantphos, K_2CO_3, *p*-ansidine; (e) B_2Pin_2, $Pd(dppf)Cl_2$, KOAc; (f) $NH_2NH_2{\cdot}H_2O$/MeOH, rt.

We have performed a series of one-step follow-up transformations to further elaborate on the synthetic utility of the activated haloalkene **3*a*/4a** scaffold (Scheme 7). First, we explored the nucleophilic addition−elimination reactions of **3a** with thiophenol and methanol in the presence of a base, which gave the corresponding products **7** and **8** in 73% yields. Next, we demonstrated typical Suzuki−Miyaura and Buchwald−Hartwig cross-coupling reactions to highlight the utility of haloalkene. The resultant coupling products **9** and **10** were obtained in 78 and 86% yields, respectively. In addition, a reaction of **3a** and B_2Pin_2 in the presence of a Pd-catalyst resulted in a homocoupled product **11**. Next, the 1,4-dicarbonyl core was successfully condensed with hydrazine hydrate, forming prominent bromopyridazine **12** with a 67% yield. This reaction provides straightforward access to bromopyridazine scaffolds, which are excellent building blocks for numerous biologically active molecules.[19]

COMPUTATIONAL STUDIES

In this computational study, we employed density functional theory (DFT) as our primary method, and calculations were performed using the Gaussian 16 software package.[20] All geometry optimization and frequency calculations were performed using the ωB97X-D[21] functional with the SMD solvation model. The 6−31+G(d,p) basis set was used for H, C, N, and O, while 6−311G(d,p) was employed for Br and I. The Gibbs free energy in solution for a given intermediate was determined as

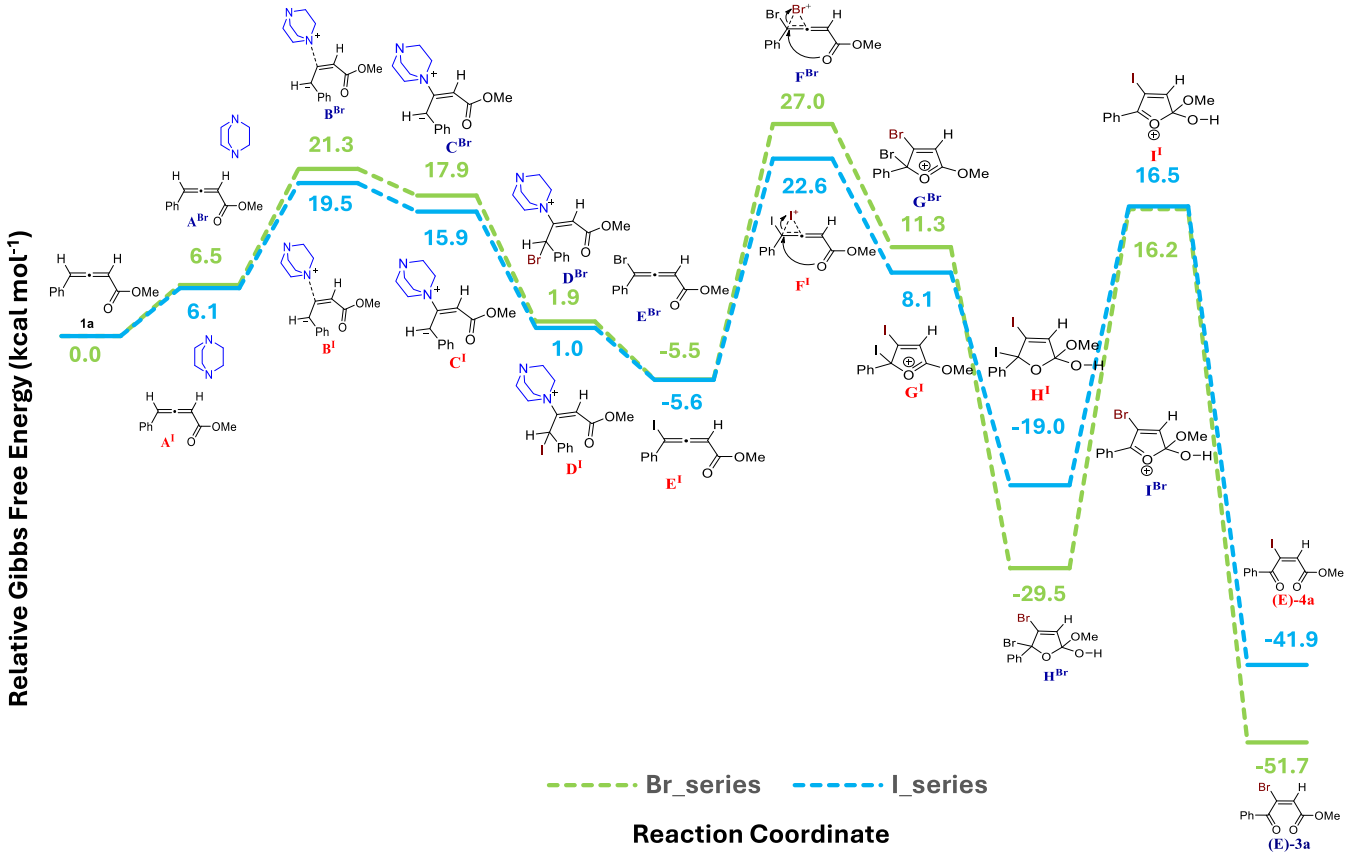
$$G_{SMD} = E_{SMD} + PV - TS_{SMD}$$

where the PV term is a constant based on the ideal gas law approximation.[22] Dichloromethane (DCM) was employed as the solvent for compounds containing bromine (Br), while acetonitrile (CH_3CN) was utilized for compounds containing iodine(I), both using the SMD solvation model.[23]

The proposed mechanism primarily results in the formation of (*E*)-**3a**/(*E*)-**4a** as the predominant products (Scheme 8). Initially, the DABCO interacts with the allenoate **1a** via van der Waals and electrostatic interactions and forms $\mathbf{A^{Br}}/\mathbf{A^{I}}$. These species proceed through transition states $\mathbf{B^{Br}}/\mathbf{B^{I}}$ at 21.3/19.5 kcal mol^{-1}, leading to allenoate-DABCO zwitterions $\mathbf{C^{Br}}/\mathbf{C^{I}}$ at 17.9/15.9 kcal mol^{-1} where the negative charge accumulates at C4 site. Consequently, $\mathbf{C^{Br}}/\mathbf{C^{I}}$ undergoes an exothermic halonium addition to yield $\mathbf{D^{Br}}/\mathbf{D^{I}}$. A subsequent proton abstraction results in haloallene $\mathbf{E^{Br}}/\mathbf{E^{I}}$ formation at −5.5/−5.6 kcal mol^{-1}. We identified an *m*/*z* peak in an in situ GC-MS analysis that corresponds to the $\mathbf{E^{Br}}$ intermediate.[13] Further, haloallenoate reacts with another equivalent of NBS/NIS to yield transient 3-membered cyclic halonium ion $\mathbf{F^{Br}}/\mathbf{F^{I}}$ that triggers five-membered cyclic oxocarbenium ion $\mathbf{G^{Br}}/\mathbf{G^{I}}$. Thereafter, it could undergo nucleophilic addition with H_2O and yield dihydrofuran $\mathbf{H^{Br}}/\mathbf{H^{I}}$ by releasing −29.5/−19.0 kcal mol^{-1} which forms $\mathbf{I^{Br}}/\mathbf{I^{I}}$ upon dehalogenation. Next, an

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Scheme 8. Proposed Mechanism Based on Experimental Results and Computational Studies



opening of cyclic oxocarbenium ion yields (*E*)-**3a**/(*E*)-**4a** at −51.7/−41.9 kcal mol^{-1}. Further, we investigated an alternative, however, a less favorable mechanism where the allenoate first reacts with NBS/NIS. The study shows that the energy of intermediates prior to the formation of $\mathbf{E^{Br}}$/$\mathbf{E^{I}}$ is notably higher (see in SI), indicating a less stable pathway compared to the mechanism in which allenoates first react with DABCO. This comparison highlights DABCO's initial interaction with the reactant and provides a more favorable pathway by stabilizing key intermediates and reducing the overall reaction energy barriers, thus enhancing reaction efficiency.

CONCLUSIONS

In summary, DABCO-promoted electrophilic bromination and iodination of allenoates and alkynoates provided a series of densely functionalized tri- and tetra-substituted activated haloalkenes in good yields. The DABCO plays a pivotal role in the demonstrated reaction, which isomerizes 3-alkynoates into allenoates and delivers a 1,4-dicarbonyl haloalkene scaffold with excellent stereoselectivity. The excellent compatibility of the optimized conditions is also demonstrated with gram-scale synthesis, highlighting the protocol's practical utility. Furthermore, a comprehensive and in-depth mechanistic elucidation was provided through computational studies and control experiments. Finally, the utility of vinyl bromide is demonstrated in cross-coupling reactions, while the 1,4-dicarbonyl functionality is explored for the synthesis of biologically active pyridazine motifs.[24] Currently, we are upgrading the protocol for the large-scale synthesis of target-oriented molecules in this laboratory.

EXPERIMENTAL SECTION

General Methods. Thin-layer chromatography analyses were performed on precoated Merck silica gel 60 F_{254} plates and the spots were visualized under a UV 254 nm light source. The crude products were purified using a Biotage automated flash chromatography instrument (Model: Selekt) on silica gel (230−400 mesh). NMR spectra were recorded in $CDCl_3$ (TMS as internal reference) and DMSO-d_6 using a 500 MHz JEOL spectrometer (JNM ECZ 500S/L1). Chemical shifts were reported as δ-values in parts per million (ppm) relative to the internal reference of tetramethylsilane (TMS). Coupling constants (J) are reported in Hertz (Hz). Multiplicities are indicated as s (singlet), d (doublet), t (triplet), q (quartet), hex (hextet), hept (heptet), dd (doublet of doublet), ddd (doublet of doublet of doublet), td (triplet of doublet), m (multiplet), br (broad singlet), qd (quartet of doublet), qt (quartet of triplet), hept (heptet), and heptd (heptet of doublet). Coupling constants (J) are reported in Hertz (Hz). IR spectra were recorded using a ThermoFisher FTIR spectrometer. Single crystal data were recorded on a Bruker D8-Venture diffractometer. Melting points were measured on Buchi Melting Point Apparatus M-565. The ESI mass spectra of the compounds were recorded on a Waters Xevo, G2-XS QTof mass spectrometer. Specific rotations were measured on a Polarimeter MCP 4100 Modular circular at multiple wavelengths. A Sigma-Aldrich Penn PhD photoreactor m2 with a blue LED lamp (450 nm) was used to irradiate the reaction mixture. The water content present in the solvents such as DCM and CH_3CN was measured on a Metrohm 870 KF Titrino plus titrator (Swiss-made). All reagents and solvents were of analytical purity, commercially available, and were used without further purification unless otherwise mentioned. All the reactions were carried out under an open atmosphere unless otherwise mentioned. Air and moisture-sensitive reactions were carried out in dried solvent under a nitrogen atmosphere using Schlenk line techniques. All the disubstituted allenoates were prepared by following a reported procedure.[10]

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*General Procedure for the Synthesis of γ-Oxo-β-bromoenoates **3a**–**3u** and **5a**–**5d**.* To oven-dried 10 mL headspace vial was added *N*-bromosuccinimide (2 equiv) and DABCO (25 mol %) in DCM (0.1 M). The solution of allenoate and alkynoate **1**:**1′** (1 equiv) in DCM (0.1 M) was added dropwise over 5 min into the reaction vial at 0 °C. The reaction mixture was allowed to stir at the same temperature until the consumption of the starting material. The solution was concentrated on a rotavapor, and the crude product was purified by flash column chromatography on silica gel. The desired product was eluted using a hexane:ethyl acetate (90:10) binary gradient.

*Procedure for Gram-Scale Synthesis of **3a**.* To an oven-dried 100 mL round-bottom flask with a magnetic stir bar was added *N*-bromosuccinimide **2a** (2.8 g, 15.6 mmol, 2.0 equiv) and DABCO (25 mol %, 219 mg) in DCM (30 mL). The solution of allenoate and alkynoate **1a**:**1a′** (1.36 g, 7.84 mmol, 1 equiv) in DCM (30 mL) was added dropwise over 10 min into the reaction mixture at 0 °C. The reaction was allowed to stir at the same temperature until the consumption of the starting material. Upon the completion of the reaction, the desired product was purified by flash column chromatography on silica gel using the binary gradient of hexane:ethyl acetate (90:10) and obtained 71% yield (1.5 g).

*Methyl (E)-3-Bromo-4-oxo-4-phenylbut-2-enoate **3a**.* The title compound was obtained by following the general procedure with 0.198 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). White solid; 44 mg (83%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.24; **m.p.** = 74–82 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.99–7.91 (m, 2H), 7.66–7.59 (m, 1H), 7.54–7.47 (m, 2H), 6.66 (s, 1H), 3.59 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 189.9, 163.5, 136.4, 134.4, 132.8, 129.4, 129.1, 125.5, 52.3; **FTIR** (ATR): ν = 1712, 1670, 1616, 1449, 1434, 1334, 1309, 1248, 1196, 1172, 1157, 1031, 1016, 992, 933, 865, 785, 728, 701, 689, 656, 622 cm^{-1}; **HRMS** (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{11}H_{10}BrO_3$ 268.9813; Found 268.9816.

*Ethyl (E)-3-Bromo-4-oxo-4-phenylbut-2-enoate **3b**.* The title compound was obtained by following the general procedure with 0.1 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (10:90). Pale yellow solid; 19 mg (67%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.43; **m.p.** = 46–50 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.95 (dd, *J* = 8.5, 1.3 Hz, 2H), 7.62 (ddt, *J* = 7.9, 7.0, 1.3 Hz, 1H), 7.54–7.46 (m, 2H), 6.65 (s, 1H), 4.02 (q, *J* = 7.1 Hz, 2H), 1.04 (t, *J* = 7.1 Hz, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 189.8, 162.9, 135.9, 134.4, 132.9, 129.5, 129.0, 126.0, 61.6, 13.7; **FTIR** (ATR): ν = 1706, 1676, 1619, 1326, 1246, 1191, 1170, 1011, 958, 893, 865, 722, 662, 601 cm^{-1}; **HRMS** (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{12}H_{12}BrO_3$ 282.9969; Found 282.9977.

*Isopropyl (E)-3-Bromo-4-oxo-4-phenylbut-2-enoate **3c**.* The title compound was obtained by following the general procedure with 0.2 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow liquid; 37 mg (63%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.44; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.98–7.95 (m, 2H), 7.62 (ddt, *J* = 7.9, 7.0, 1.3 Hz, 1H), 7.53–7.48 (m, 2H), 6.63 (s, 1H), 4.87 (hept, *J* = 6.3 Hz, 1H), 1.00 (d, *J* = 6.2 Hz, 6H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 189.8, 162.4, 135.4, 134.3, 133.0, 129.5, 129.0, 126.5, 69.6, 21.3; **FTIR** (ATR): ν = 2990, 1712, 1677, 1615, 1449, 1321, 1266, 1222, 1193, 1173, 1102, 1033, 1018, 971, 915, 869, 822, 700, 685, 656 cm^{-1}; **HRMS** (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{13}H_{14}BrO_3$ 297.0126; Found 297.0126.

*2-(4-Isobutylphenyl)propyl (E)-3-Bromo-4-oxo-4-phenylbut-2-enoate **3d**.* The title compound was obtained by following the general procedure with 0.13 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (1:9). Pale yellow liquid; 36.5 mg (65%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.33; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.94–7.92 (m, 2H), 7.62 (ddt, *J* = 7.8, 7.0, 1.3 Hz, 1H), 7.51–7.47 (m, 2H), 7.03 (d, *J* = 8.2 Hz, 2H), 6.97 (d, *J* = 8.0 Hz, 2H), 6.63 (s, 1H), 4.09 (dd, *J* = 10.9, 6.3 Hz, 1H), 3.99 (dd, *J* = 10.9, 7.8 Hz, 1H), 2.83 (h, *J* = 7.0 Hz, 1H), 2.42 (d, *J* = 7.2 Hz, 2H), 1.82 (dt, *J* = 13.3, 6.7 Hz, 1H), 1.10 (d, *J* = 7.0 Hz, 3H), 0.88 (d, *J* = 6.6 Hz, 6H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 189.8, 163.0, 140.2, 139.7, 136.1, 134.4, 132.8, 129.6, 129.3, 129.1, 126.9, 125.8, 70.7, 45.1, 38.3, 30.3, 22.5, 18.1; **FTIR** (ATR): ν = 1716, 1680, 1449, 1326, 1238, 1224, 1185, 1169, 1018, 989, 866, 798, 714, 685, 654 cm^{-1}; **HRMS** (ESI) *m*/*z*: $[M + Na]^+$ calcd for $C_{23}H_{25}BrO_3Na$ 451.0884; Found 451.0886.

*(Z)-Octadec-9-en-1-yl (E)-3-Bromo-4-oxo-4-phenylbut-2-enoate **3e**.* The title compound was obtained by following the general procedure with 0.23 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow liquid; 73 mg (63%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.42; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.96–7.91 (m, 2H), 7.63–7.58 (m, 1H), 7.49 (dd, *J* = 8.2, 7.4 Hz, 2H), 6.63 (s, 1H), 4.23–4.17 (m, 2H), 3.94 (t, *J* = 6.6 Hz, 2H), 2.01 (dtd, *J* = 16.0, 6.2, 2.9 Hz, 2H), 1.87–1.76 (m, 2H), 1.61–1.50 (m, 2H), 1.37–1.20 (m, 22H), 0.86 (d, *J* = 7.1 Hz, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 189.7, 163.0, 135.8, 134.4, 132.9, 129.5, 129.0, 126.0, 65.8, 59.93, 59.87, 35.0, 34.97, 31.9, 29.5, 29.3, 29.2, 29.1, 28.9, 28.8, 28.2, 27.91, 27.86, 25.7, 22.8, 14.2; **FTIR** (ATR): ν = 2922, 2853, 1717, 1682, 1449, 1327, 1251, 1223, 1187, 1171, 1033, 1018, 865, 719, 701, 685, 655, 540 cm^{-1}; **HRMS** (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{28}H_{42}BrO_3$ 505.2317; Found 505.2322.

*Methyl (E)-3-Bromo-4-oxo-4-(p-tolyl)but-2-enoate **3f**.* The title compound was obtained by following the general procedure with 0.2 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 41.5 mg (73%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.31; **m.p.** = 82–85 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.82 (d, *J* = 8.2 Hz, 2H), 7.30–7.26 (m, 2H), 6.62 (s, 1H), 3.57 (s, 3H), 2.41 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 189.6, 163.6, 145.6, 136.6, 130.4, 129.8, 129.6, 125.3, 52.3, 22.0; **FTIR** (ATR): ν = 1719, 1672, 1617, 1601, 1432, 1331, 1297, 1242, 1170, 1000, 823, 726, 638, 611 cm^{-1}; **HRMS** (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{12}H_{12}BrO_3$ 282.9969; Found 282.9970.

*Methyl (E)-3-Bromo-4-(4-(tert-butyl)phenyl)-4-oxobut-2-enoate **3g**.* The title compound was obtained by following the general procedure with 0.16 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow liquid; 33.5 mg (64%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.36; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.86 (d, *J* = 8.6 Hz, 2H), 7.50 (d, *J* = 8.6 Hz, 2H), 6.63 (s, 1H), 3.58 (s, 3H), 1.33 (s, 9H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 189.6, 163.6, 158.4, 136.8, 130.2, 129.4, 126.1, 125.2, 52.3, 35.4, 31.1; **FTIR** (ATR): ν = 2976, 1721, 1674, 1616, 1602, 1330, 1250, 1173, 1027, 997, 858, 637, 620 cm^{-1}; **HRMS** (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{15}H_{18}BrO_3$ 325.0439; Found 325.0445.

*Methyl (E)-3-Bromo-4-(4-fluorophenyl)-4-oxobut-2-enoate **3h**.* The title compound was obtained by following the general procedure with 0.23 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (10:90). Brownish solid; 45 mg (67%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.27; **m.p.** = 60–63 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.95 (dd, *J* = 8.9, 5.3 Hz, 2H), 7.16 (dd, *J* = 8.9, 8.3 Hz, 2H), 6.64 (s, 1H), 3.59 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 188.4, 166.5 (d, *J* = 257.1 Hz), 163.5, 136.0, 132.2 (d, *J* = 9.7 Hz), 129.3 (d, *J* = 3.0 Hz), 125.6, 116.4 (d, *J* = 22.3 Hz), 52.4; **^{19}F NMR** (471 MHz, $CDCl_3$) δ = −102.1; **FTIR** (ATR): ν = 1723, 1674, 1591, 1436, 1332, 1233, 1197, 1154, 1027, 978, 936, 857, 737, 621, 611 cm^{-1}; **HRMS** (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{11}H_9BrFO_3$ 286.9719; Found 286.9719.

*Methyl (E)-3-Bromo-4-(4-chlorophenyl)-4-oxobut-2-enoate **3i**.* The title compound was obtained by following the general procedure with 0.23 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 37.5 mg (62%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.43; **m.p.** = 100–107 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.86 (d, *J* = 8.7 Hz, 2H), 7.46 (d, *J* = 8.6 Hz, 2H), 6.64 (s, 1H), 3.59 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$)

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δ = 188.7, 163.5, 141.0, 135.8, 131.3, 130.8, 129.5, 125.7, 52.4; **FTIR** (ATR): ν = 1717, 1673, 1615, 1584, 1433, 1331, 1197, 1110, 1025, 980, 964, 860, 756, 631, 619 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9BrClO_3$ 302.9423; Found 302.9423.

Methyl (E)-3-Bromo-4-(4-bromophenyl)-4-oxobut-2-enoate ***3j.*** The title compound was obtained by following the general procedure with 0.18 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (10:90). Yellow solid; 42.5 mg (68%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.3; **m.p.** = 90−93 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 7.78 (d, J = 8.6 Hz, 2H), 7.63 (d, J = 8.6 Hz, 2H), 6.64 (s, 1H), 3.59 (s, 3H); ^{13}**C** {^{1}**H**} **NMR** (126 MHz, $CDCl_3$) δ = 188.9, 163.5, 135.7, 132.5, 131.7, 130.8, 129.8, 125.8, 52.4; **FTIR** (ATR): ν = 1717, 1674, 1616, 1581, 1569, 1433, 1333, 1250, 1235, 1199, 1172, 1068, 1008, 996, 934, 862, 823, 748, 665, 631, 618 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9Br_2O_3$ 346.8918; Found 346.8923.

Methyl (E)-3-Bromo-4-(2-methoxyphenyl)-4-oxobut-2-enoate ***3k.*** The title compound was obtained by following the general procedure with 0.07 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (18:82). Yellow solid; 14 mg (67%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.16; **m.p.** = 90−95 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 7.96 (dd, J = 7.9, 1.8 Hz, 1H), 7.54 (ddt, J = 9.4, 7.4, 1.3 Hz, 1H), 7.09−7.01 (m, 1H), 6.97 (d, J = 8.4 Hz, 1H), 6.39 (d, J = 0.5 Hz, 1H), 3.86 (s, 3H), 3.57 (s, 3H); ^{13}**C** {^{1}**H**} **NMR** (126 MHz, $CDCl_3$) δ = 188.2, 164.0, 159.9, 139.8, 135.8, 132.1, 123.0, 122.1, 121.1, 112.4, 55.8, 52.1; **FTIR** (ATR): ν = 2922, 1708, 1645, 1618, 1592, 1487, 1332, 1292, 1163, 1054, 995, 935, 778, 654, 582 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{12}BrO_4$ 298.9919; Found 298.9920.

Methyl (E)-3-Bromo-4-(2-fluorophenyl)-4-oxobut-2-enoate ***3l.*** The title compound was obtained by following the general procedure with 0.155 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). White solid; 35 mg (79%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.18; **m.p.** = 41−44 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 7.93 (dddd, J = 7.8, 7.3, 1.9, 0.3 Hz, 1H), 7.53 (dddd, J = 8.3, 7.3, 5.0, 1.8 Hz, 1H), 7.23 (ddd, J = 7.9, 7.3, 1.1 Hz, 1H), 7.11−7.05 (m, 1H), 6.45 (d, J = 1.1 Hz, 1H), 3.55 (s, 3H); ^{13}**C** {^{1}**H**} **NMR** (126 MHz, $CDCl_3$) δ = 186.3, 163.9, 162.2 (d, J = 260.0 Hz), 137.7, 136.3 (d, J = 9.2 Hz), 131.6, 124.8 (d, J = 3.6 Hz), 124.0 (d, J = 3.0 Hz), 121.9 (d, J = 9.3 Hz), 117.0 (d, J = 22.1 Hz), 52.3; ^{19}**F NMR** (471 MHz, $CDCl_3$) δ = −108.7; **FTIR** (ATR): ν = 1707, 1683, 1664, 1604, 1478, 1434, 1333, 1233, 1194, 1146, 1019, 934, 849, 748, 710, 555 cm^{-1}; **HRMS** (ESI) m/z: $[M + Na]^+$ calcd for $C_{11}H_8BrFO_3Na$ 308.9538; Found 308.9537.

Methyl (E)-3-Bromo-4-oxo-4-(m-tolyl)but-2-enoate ***3m.*** The title compound was obtained by following the general procedure with 0.18 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 28.5 mg (56%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.4; **m.p.** = 50−60 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 7.76−7.58 (m, 2H), 7.41−7.26 (m, 2H), 6.58 (s, 1H), 3.53 (s, 3H), 2.35 (s, 3H); ^{13}**C** {^{1}**H**} **NMR** (126 MHz, $CDCl_3$) δ = 190.0, 163.6, 139.0, 136.6, 135.3, 132.8, 129.7, 128.9, 126.9, 125.3, 52.3, 21.4; **FTIR** (ATR): ν = 1707, 1670, 1617, 1600, 1431, 1328, 1263, 1209, 1157, 1051, 988, 944, 870, 720, 650 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{12}BrO_3$ 282.9969; Found 282.9970.

Methyl (E)-3-Bromo-4-(3-methoxyphenyl)-4-oxobut-2-enoate ***3n.*** The title compound was obtained by following the general procedure with 0.22 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow liquid; 41 mg (62%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.27; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 7.44 (dd, J = 2.7, 1.6 Hz, 1H), 7.41 (ddd, J = 7.6, 1.6, 1.0 Hz, 1H), 7.36−7.31 (m, 1H), 7.10 (ddd, J = 8.2, 2.7, 1.0 Hz, 1H), 6.59 (s, 1H), 3.81 (s, 3H), 3.54 (s, 3H); ^{13}**C** {^{1}**H**} **NMR** (126 MHz, $CDCl_3$) δ = 189.8, 163.6, 160.2, 136.3, 134.1, 130.1, 125.5, 122.4, 121.3, 112.9, 55.6, 52.4; **FTIR** (ATR): ν = 1717, 1674, 1618, 1594, 1581, 1433, 1329, 1220, 1155, 1027, 997, 858, 736, 696, 637 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{12}BrO_4$ 298.9918; Found 298.9919.

Methyl (E)-3-Bromo-4-(3-fluorophenyl)-4-oxobut-2-enoate ***3o.*** The title compound was obtained by following the general procedure with 0.165 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (10:90). Yellow solid; 27 mg (57%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.37; **m.p.** = 50−55 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 7.67−7.61 (m, 1H), 7.57 (ddd, J = 9.0, 2.7, 1.6 Hz, 1H), 7.42 (td, J = 8.0, 5.4 Hz, 1H), 7.25 (tdd, J = 8.3, 2.7, 1.0 Hz, 1H), 6.60 (s, 1H), 3.54 (s, 3H); ^{13}**C** {^{1}**H**} **NMR** (126 MHz, $CDCl_3$) δ = 188.7, 163.5, 163.0 (d, J = 252 Hz), 135.6, 135.0 (d, J = 6.5 Hz), 130.8 (d, J = 7.7 Hz) 125.9, 125.3 (d, J = 3.1 Hz), 121.5 (d, J = 21.5), 115.9 (d, J = 22.8 Hz), 52.4; ^{19}**F NMR** (471 MHz, $CDCl_3$) δ = −110.9; **FTIR** (ATR): ν = 1709, 1676, 1626, 1588, 1430, 1336, 1209, 1145, 1001, 876, 787, 721, 672 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9BrFO_3$ 286.9719; Found 286.9732.

Methyl (E)-3-Bromo-4-(3-chlorophenyl)-4-oxobut-2-enoate ***3p.*** The title compound was obtained by following the general procedure with 0.14 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). White solid; 18 mg (42%); **TLC** (SiO_2; EtOAc:hexanes 1:4 UV) R_f = 0.57; **m.p.** = 59−68 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 7.90 (ddd, J = 2.1, 1.6, 0.4 Hz, 1H), 7.80 (ddd, J = 7.8, 1.7, 1.1 Hz, 1H), 7.58 (ddd, J = 8.0, 2.1, 1.1 Hz, 1H), 7.44 (td, J = 7.9, 0.4 Hz, 1H), 6.66 (s, 1H), 3.61 (s, 3H); ^{13}**C** {^{1}**H**} **NMR** (126 MHz, $CDCl_3$) δ = 188.6, 163.6, 135.5, 135.4, 134.5, 134.3, 130.4, 129.2, 127.5, 125.9, 52.5; **FTIR** (ATR): ν = 1709, 1682, 1620, 1572, 1426, 1329, 1237, 1173, 1076, 1042, 945, 851, 765, 687, 623 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9BrClO_3$ 302.9423; Found 302.9424.

Methyl (E)-3-Bromo-4-(3-bromophenyl)-4-oxobut-2-enoate ***3q.*** The title compound was obtained by following the general procedure with 0.16 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:exanes (10:90). Yellow solid; 24 mg (43%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.4; **m.p.** = 80−85 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 8.00 (td, J = 1.8, 0.4 Hz, 1H), 7.78 (ddd, J = 7.8, 1.6, 1.0 Hz, 1H), 7.67 (ddd, J = 7.9, 2.0, 1.0 Hz, 1H), 7.32 (td, J = 7.9, 0.4 Hz, 1H), 6.60 (s, 1H), 3.55 (s, 3H); ^{13}**C** {^{1}**H**} **NMR** (126 MHz, $CDCl_3$) δ = 188.5, 163.6, 137.2, 135.4, 134.6, 132.1, 130.6, 128.0, 125.9, 123.4, 52.5; **FTIR** (ATR): ν = 1719, 1681, 1621, 1563, 1424, 1333, 1241, 1178, 1002, 942, 863, 680, 633 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9Br_2O_3$ 346.8918; Found 346.8911.

Methyl (E)-3-Bromo-4-(naphthalen-1-yl)-4-oxobut-2-enoate ***3r.*** The title compound was obtained by following the general procedure with 0.22 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). White solid; 36.5 mg (52%); **TLC** (SiO_2; EtOAc:hexanes 5:95 UV) R_f = 0.45; **m.p.** = 105−112 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 9.16 (dd, J = 8.7, 1.0 Hz, 1H), 8.07 (dt, J = 8.2, 0.7 Hz, 1H), 8.01 (dd, J = 7.3, 1.2 Hz, 1H), 7.90 (ddd, J = 8.2, 1.4, 0.7 Hz, 1H), 7.70 (ddd, J = 8.6, 6.8, 1.4 Hz, 1H), 7.58 (ddd, J = 8.1, 6.8, 1.2 Hz, 1H), 7.50 (dd, J = 8.2, 7.3 Hz, 1H), 6.66 (d, J = 0.5 Hz, 1H), 3.48 (s, 3H); ^{13}**C NMR** (126 MHz, $CDCl_3$) δ = 191.2, 163.6, 137.7, 135.4, 134.2, 132.7, 131.3, 129.3, 129.2, 128.7, 127.0, 126.2, 125.1, 124.4, 52.4; **FTIR** (ATR): ν = 1706, 1655, 1611, 1326, 1238, 1207, 1102, 1074, 980, 923, 868, 794, 777, 652, 642, 566 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{12}BrO_3$ 318.9970; Found 318.9973.

Methyl (E)-3-Bromo-4-(naphthalen-2-yl)-4-oxobut-2-enoate ***3s.*** The title compound was obtained by following the general procedure with 0.21 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 41 mg (61%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.33; **m.p.** = 89−93 °C; ^{1}**H NMR** (500 MHz, $CDCl_3$) δ = 8.43−8.38 (m, 1H), 8.03 (dd, J = 8.6, 1.7 Hz, 1H), 7.98−7.91 (m, 2H), 7.88 (dd, J = 8.3, 1.1 Hz, 1H), 7.62 (ddd, J = 8.2, 6.9, 1.3 Hz, 1H), 7.55 (ddd, J = 8.2, 6.9, 1.3 Hz, 1H), 6.72 (s, 1H),

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3.56 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 189.9, 163.6, 136.5, 136.3, 132.6, 132.1, 130.3, 129.9, 129.3, 129.2, 128.1, 127.1, 125.7, 124.1, 52.4; FTIR (ATR): ν = 1704, 1666, 1621, 1596, 1427, 1353, 1337, 1273, 1170, 1029, 957, 881, 866, 770, 715, 643, 590 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{12}BrO_3$ 318.9969; Found 318.9970.

Methyl (E)-3-Bromo-4-(cyclohex-1-en-1-yl)-4-oxobut-2-enoate ***3t***. The title compound was obtained by following the general procedure with 0.24 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Colorless liquid; 11 mg (17%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.63; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 6.79 (tt, J = 3.9, 1.6 Hz, 1H), 6.43 (s, 1H), 3.61 (s, 3H), 2.26 (tt, J = 6.1, 2.2 Hz, 2H), 2.22 (tdd, J = 6.0, 3.9, 2.4 Hz, 2H), 1.65−1.57 (m, 4H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.3, 163.6, 145.7, 137.0, 136.4, 124.9, 52.3, 26.4, 22.6, 21.6, 21.5; FTIR (ATR): ν = 1716, 1661, 1633, 1616, 1434, 1327, 1269, 1228, 1191, 1169, 996, 960, 930, 905, 863, 684, 623 cm^{-1}; HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{11}H_{13}BrO_3Na$ 294.9945; Found 294.9953.

Methyl (E)-3-Bromo-4-(cyclohept-1-en-1-yl)-4-oxobut-2-enoate ***3u***. The title compound was obtained by following the general procedure with 0.225 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow liquid; 35 mg (54%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.29; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 6.99−6.92 (m, 1H), 6.42 (s, 1H), 3.61 (s, 3H), 2.54−2.49 (m, 2H), 2.35−2.29 (m, 2H), 1.77−1.70 (m, 2H), 1.53−1.45 (m, 4H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.4, 163.5, 150.7, 142.6, 137.4, 124.7, 52.2, 32.0, 29.6, 25.8, 25.7, 25.3; FTIR (ATR): ν = 1720, 1645, 1630, 1415, 1296, 1252, 1215, 1170, 1110, 980, 880, 720, 670 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{16}BrO_3$ 287.0283; Found 287.0271.

General Procedure for the Synthesis of γ-Oxo-β-iodoenoates ***4a−4u*** *and* ***6a−6b***. To an oven-dried 10 mL headspace vial was added *N*-iodosuccinimide **2e** (2.5 equiv) and DABCO (25 mol %) in CH_3CN (0.1 M). A solution of allenoates and alkynoates **1**:**1′** (1 equiv) in CH_3CN (0.1 M) was added dropwise to the reaction mixture at 25 °C for 5 min. The solution was allowed to stir at the same temperature until the consumption of the starting material. After the consumption of the starting material, the reaction mixture was concentrated on a rotavapor, and the crude product was purified by flash column chromatography. The desired product **4**/**6** was eluted using hexane:ethyl acetate (90:10) binary gradient.

Procedure for Gram-Scale Synthesis of ***4a***. To an oven-dried 100 mL round-bottom flask with a magnetic stir bar was added *N*-iodosuccinimide **2e** (2.88 g, 15.6 mmol, 2.0 equiv) and DABCO (25 mol %, 179 mg) in CH_3CN (30 mL). The solution of allenoate and alkynoate **1a**:**1a′** (1.36 g, 7.84 mmol, 1 equiv) in CH_3CN (30 mL) was added dropwise over 5 min into the reaction mixture at 25 °C. The reaction was allowed to stir at the same temperature until the consumption of the starting material. Upon the completion of the reaction, the desired product was purified by flash column chromatography on silica gel using the binary gradient of hexane:ethyl acetate (90:10) and obtained a 67% yield (1.35 g).

Methyl (E)-3-Iodo-4-oxo-4-phenylbut-2-enoate ***4a***. The title compound was obtained by following the general procedure with 0.2 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). White solid; 47.5 mg (75%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.23; m.p. = 80−88 °C; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 7.97−7.92 (m, 2H), 7.64−7.58 (m, 1H), 7.52−7.47 (m, 2H), 6.86 (s, 1H), 3.57 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.9, 163.5, 134.2, 133.1, 132.1, 129.4, 129.0, 114.5, 52.3; FTIR (ATR): ν = 1715, 1668, 1596, 1449, 1434, 1330, 1307, 1235, 1197, 1172, 1050, 930, 723, 686, 603 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_{10}IO_3$ 316.9674; Found 316.9675.

Ethyl (E)-3-Iodo-4-oxo-4-phenylbut-2-enoate ***4b***. The title compound was obtained by following the general procedure with 0.22 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow liquid; 52.5 mg (73%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.33; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 7.90−7.85 (m, 2H), 7.56−7.51 (m, 1H), 7.42 (dddd, J = 8.5, 6.2, 1.4, 0.9 Hz, 2H), 6.78 (s, 1H), 3.93 (q, J = 7.1 Hz, 2H), 0.96 (t, J = 7.1 Hz, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.8, 163.0, 134.2, 133.5, 132.2, 129.5, 129.0, 114.1, 61.5, 13.7; FTIR (ATR): ν = 1712, 1671, 1598, 1448, 1366, 1324, 1235, 1223, 1189, 1172, 1025, 1010, 949, 867, 839, 771, 714, 685, 645, 608 cm^{-1}; HRMS (ESI) m/z: $[M]^+$ calcd for $C_{12}H_{11}IO_3$ 329.9752; Found 329.9757.

Isopropyl (E)-3-Iodo-4-oxo-4-phenylbut-2-enoate ***4c***. The title compound was obtained by following the general procedure with 0.245 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow liquid; 59 mg (70%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.41; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 7.90−7.87 (m, 2H), 7.54 (ddt, J = 7.9, 6.9, 1.3 Hz, 1H), 7.46−7.38 (m, 2H), 6.76 (s, 1H), 4.78 (p, J = 6.3 Hz, 1H), 0.92 (d, J = 6.2 Hz, 6H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.7, 162.4, 134.2, 134.0, 132.3, 129.5, 129.0, 113.6, 69.6, 21.4; FTIR (ATR): ν = 1708, 1670, 1597, 1449, 1319, 1263, 1238, 1221, 1172, 1193, 1144, 1101, 1031, 967, 865, 818, 685 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{13}H_{14}IO_3$ 344.9987; Found 344.9988.

2-(4-Isobutylphenyl)propyl (E)-3-Iodo-4-oxo-4-phenylbut-2-enoate ***4d***. The title compound was obtained by following the general procedure with 0.195 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow liquid; 59 mg (63%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.25; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 7.96−7.90 (m, 2H), 7.63−7.57 (m, 1H), 7.48 (t, J = 7.8 Hz, 2H), 7.04 (d, J = 8.2 Hz, 2H), 6.98 (d, J = 8.1 Hz, 2H), 6.83 (s, 1H), 4.07 (dd, J = 10.9, 6.3 Hz, 1H), 3.98 (dd, J = 10.9, 7.9 Hz, 1H), 2.83 (h, J = 7.1 Hz, 1H), 2.42 (d, J = 7.2 Hz, 2H), 1.83 (dp, J = 13.6, 6.8 Hz, 1H), 1.10 (d, J = 7.0 Hz, 3H), 0.88 (d, J = 6.6 Hz, 6H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.8, 163.0, 140.2, 139.8, 134.2, 133.4, 132.1, 129.5, 129.3, 129.1, 126.9, 114.3, 70.6, 45.1, 38.3, 30.3, 22.5, 18.1; FTIR (ATR): ν = 2950, 1715, 1674, 1598, 1448, 1323, 1233, 1183, 1169, 1032, 1017, 986, 949, 866, 844, 798, 772, 711, 685, 644, 609 cm^{-1}; HRMS (ESI) m/z: $[M+K]^+$ calcd for $C_{23}H_{25}IO_3K$ 515.0485; Found 515.0481.

(Z)-Octadec-9-en-1-yl (E)-3-Iodo-4-oxo-4-phenylbut-2-enoate ***4e***. The title compound was obtained by following the general procedure with 0.10 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow liquid; 28 mg (51%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.33; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 7.88 (dd, J = 8.4, 1.3 Hz, 2H), 7.55−7.52 (m, 1H), 7.44−7.40 (m, 2H), 6.78 (s, 1H), 5.33−5.23 (m, 2H), 3.87 (t, J = 6.6 Hz, 2H), 1.96−1.92 (m, 3H), 1.30−1.10 (m, 25H), 0.82−0.79 (m, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.7, 163.1, 134.2, 133.6, 132.1, 130.1, 129.9, 129.5, 129.0, 114.0, 65.8, 32.0, 29.9, 29.87, 29.82, 29.6, 29.4, 29.3, 29.28, 28.19, 27.32, 27.28, 25.8, 22.8, 14.2; FTIR (ATR): ν = 2924, 2854, 1716, 1675, 1599, 1449, 1324, 1222, 1186, 1171, 1048, 1015, 888, 864, 773, 716, 686 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{28}H_{42}IO_3$ 553.2178; Found 553.2179.

Methyl (E)-3-Iodo-4-oxo-4-(p-tolyl)but-2-enoate ***4f***. The title compound was obtained by following the general procedure with 0.216 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (10:90). Yellow solid; 40.4 mg (57%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.29; m.p. = 87−92 °C; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 7.84 (d, J = 8.2 Hz, 2H), 7.29 (d, J = 7.6 Hz, 2H), 6.84 (s, 1H), 3.57 (s, 3H), 2.43 (s, 3H); ^{13}C NMR (126 MHz, $CDCl_3$) δ = 191.6, 163.5, 145.4, 132.8, 129.8, 129.6, 114.9, 52.3, 22.0; FTIR (ATR): ν = 1716, 1667, 1599, 1431, 1327, 1239, 1195, 1182, 1168, 1114, 1025, 992, 931, 852, 820, 789, 721, 690, 683, 601 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{12}IO_3$ 330.9831; Found 330.9843.

Methyl (E)-4-(4-(tert-Butyl)phenyl)-3-iodo-4-oxobut-2-enoate ***4g***. The title compound was obtained by following the general procedure with 0.26 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using

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eluent EtOAc:hexanes (5:95). Yellow waxy liquid; 72.5 mg (75%); **TLC** (SiO_2; EtOAc:hexanes 5:95 UV) R_f = 0.15; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.85 (d, J = 8.6 Hz, 2H), 7.48 (d, J = 8.7 Hz, 2H), 6.83 (s, 1H), 3.56 (s, 3H), 1.32 (s, 9H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 191.6, 163.6, 158.1, 132.8, 129.4, 126.1, 115.0, 52.3, 35.4, 31.1; **FTIR** (ATR): ν = 1706, 1664, 1599, 1320, 1266, 1227, 1163, 1107, 1027, 992, 930, 863, 816, 671, 602 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{18}IO_3$ 373.0301; Found 373.0301.

Methyl (E)-4-(4-Fluorophenyl)-3-iodo-4-oxobut-2-enoate ***4h***. The title compound was obtained by following the general procedure with 0.245 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow liquid; 54.5 mg (66%); **TLC** (SiO_2; EtOAc:hexanes 5:95 UV) R_f = 0.12; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.95 (dd, J = 9.0, 5.3 Hz, 2H), 7.15 (dd, J = 9.0, 8.3 Hz, 2H), 6.83 (s, 1H), 3.58 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 190.5, 166.4 (d, J = 256.8 Hz), 163.5, 133.1, 132.1 (d, J = 9.7 Hz), 128.5 (d, J = 3.1 Hz), 116.4 (d, J = 22.2 Hz), 114.0, 52.3; **^{19}F NMR** (471 MHz, $CDCl_3$) δ = −102.8; **FTIR** (ATR): ν = 1715, 1671, 1592, 1504, 1504, 1327, 1235, 1195, 1176, 1150, 1024, 993, 930, 856, 837, 805 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9FIO_3$ 334.9580; Found 334.9580.

Methyl (E)-4-(4-Chlorophenyl)-3-iodo-4-oxobut-2-enoate ***4i***. The title compound was obtained by following the general procedure with 0.24 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 52 mg (62%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.33; **m.p.** = 88−99 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.86 (d, J = 8.6 Hz, 2H), 7.45 (d, J = 8.6 Hz, 2H), 6.84 (s, 1H), 3.58 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 190.8, 163.6, 140.7, 133.3, 130.7, 130.5, 129.5, 113.8, 52.4; **FTIR** (ATR): ν = 1715, 1668, 1603, 1583, 1568, 1431, 1327, 1234, 1196, 1169, 1091, 1022, 1009, 989, 930, 858, 818, 683, 601 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9ICIO_3$ 350.9285; Found 350.9285.

Methyl (E)-4-(4-Bromophenyl)-3-iodo-4-oxobut-2-enoate ***4j***. The title compound was obtained by following the general procedure with 0.2 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (7:93). White solid; 55 mg (70%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.32; **m.p.** = 104−108 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.80 (d, J = 8.7 Hz, 2H), 7.64 (d, J = 8.6 Hz, 2H), 6.86 (s, 1H), 3.60 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 190.9, 163.5, 133.3, 132.4, 130.7, 129.5, 113.7, 52.3; **FTIR** (ATR): ν = 1713, 1666, 1610, 1579, 1567, 1432, 1397, 1249, 1240, 1197, 1068, 1050, 996, 932, 592, 554 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9BrIO_3$ 394.8779; Found 394.8758.

Methyl (E)-4-(2-Fluorophenyl)-3-iodo-4-oxobut-2-enoate ***4k***. The title compound was obtained by following the general procedure with 0.125 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow liquid; 25 mg (60%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.4; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 8.00 (td, J = 7.6, 1.9 Hz, 1H), 7.61−7.53 (m, 1H), 7.29−7.25 (m, 1H), 7.13 (ddd, J = 11.1, 8.3, 1.1 Hz, 1H), 6.67 (d, J = 1.1 Hz, 1H), 3.59 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 188.5, 163.8, 162.0 (d, J = 259.1 Hz), 136.1 (d, J = 9.2 Hz), 131.6, 131.5 (d, J = 3.0 Hz), 124.8 (d, J = 3.6 Hz), 121.1 (d, J = 9.6 Hz), 117.0 (d, J = 22.3 Hz), 115.9, 52.3; **^{19}F NMR** (471 MHz, $CDCl_3$) δ = −107.3; **FTIR** (ATR): ν = 1712, 1662, 1606, 1480, 1452, 1433, 1327, 1275, 1196, 1177, 1045, 993, 864, 765, 639, 607 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9FIO_3$ 334.9580; Found 334.9581.

Methyl (E)-3-Iodo-4-(2-methoxyphenyl)-4-oxobut-2-enoate ***4l***. The title compound was obtained by following the general procedure with 0.17 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Brown solid; 34 mg (57%); **TLC** (SiO_2; EtOAc:hexanes 95:5 UV) R_f = 0.11; **m.p.** = 127−135 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.98 (dd, J = 7.8, 1.8 Hz, 1H), 7.57−7.48 (m, 1H), 7.05 (ddd, J = 8.0, 7.3, 1.0 Hz, 1H), 6.97 (dd, J = 8.4, 1.0 Hz, 1H), 6.55 (s, 1H), 3.86 (s, 3H), 3.55 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 190.3, 164.0, 159.7, 135.7, 132.1, 129.4, 122.4, 121.1, 118.6, 112.3, 55.6, 52.1; **FTIR** (ATR): ν = 1700, 1637, 1607, 1593, 1486, 1466, 1429, 1332, 1290, 1250, 1195, 1167, 1122, 1016, 996, 862, 776, 644, 612, 578 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{12}IO_4$ 346.9780; Found 346.9780.

Methyl (E)-3-Iodo-4-oxo-4-(m-tolyl)but-2-enoate ***4m***. The title compound was obtained by following the general procedure with 0.19 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). White solid; 42 mg (67%); **TLC** (SiO_2; EtOAc:hexanes 5:95 UV) R_f = 0.16; **m.p.** = 66−70 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.77−7.74 (m, 1H), 7.70 (ddd, J = 7.5, 1.2, 0.6 Hz, 1H), 7.41−7.38 (m, 1H), 7.35 (t, J = 7.6 Hz, 1H), 6.83 (s, 1H), 3.56 (s, 3H), 2.40 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 192.1, 163.6, 139.0, 135.1, 132.9, 132.1, 129.7, 128.9, 126.8, 114.8, 52.3, 21.5; **FTIR** (ATR): ν = 1704, 1667, 1600, 1581, 1431, 1330, 1260, 1213, 1170, 1031, 996, 941, 874, 828, 789, 680 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{12}IO_3$ 330.9831; Found 330.9845.

Methyl (E)-3-Iodo-4-(3-methoxyphenyl)-4-oxobut-2-enoate ***4n***. The title compound was obtained by following the general procedure with 0.14 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow liquid; 29.5 mg (60%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.14; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.49 (dd, J = 2.7, 1.5 Hz, 1H), 7.46 (ddd, J = 7.6, 1.6, 1.0 Hz, 1H), 7.37 (ddd, J = 8.1, 7.6, 0.4 Hz, 1H), 7.13 (ddd, J = 8.2, 2.7, 1.0 Hz, 1H), 6.83 (s, 1H), 3.85 (s, 3H), 3.57 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 191.7, 163.6, 160.2, 133.4, 133.0, 130.0, 122.4, 121.1, 114.4, 112.9, 55.6, 52.3; **FTIR** (ATR): ν = 1715, 1676, 1599, 1434, 1273, 1238, 1177, 1105, 1069, 1019, 997, 960, 909, 787, 781, 726, 686, 658, 630, 599 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{12}IO_4$ 346.9780; Found 346.9781.

Methyl (E)-4-(3-Fluorophenyl)-3-iodo-4-oxobut-2-enoate ***4o***. The title compound was obtained by following the general procedure with 0.18 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 44.5 mg (74%); **TLC** (SiO_2; EtOAc:hexanes 5:95 UV) R_f = 0.13; **m.p.** = 70−77 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.69 (ddd, J = 7.7, 1.6, 1.0 Hz, 1H), 7.62 (ddd, J = 9.1, 2.6, 1.6 Hz, 1H), 7.46 (td, J = 8.0, 5.4 Hz, 1H), 7.29 (tdd, J = 8.2, 2.6, 1.0 Hz, 1H), 6.85 (s, 1H), 3.58 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 190.7, 163.5, 163.0 (d, J = 249 Hz), 134.2 (d, J = 6.6 Hz), 133.4, 130.7 (d, J = 7.7 Hz), 125.2 (d, J = 3.0 Hz), 121.3 (d, J = 21.6 Hz), 115.9 (d, J = 22.5 Hz), 113.6, 52.4; **^{19}F NMR** (471 MHz, $CDCl_3$) δ = −111.0; **FTIR** (ATR): ν = 1708, 1662, 1603, 1587, 1443, 1433, 1326, 1258, 1206, 1172, 1145, 998, 948, 864, 788, 718, 672, 642, 620 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9IFO_3$ 334.9580; Found 334.9583.

Methyl (E)-4-(3-Chlorophenyl)-3-iodo-4-oxobut-2-enoate ***4p***. The title compound was obtained by following the general procedure with 0.257 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 59 mg (65%); **TLC** (SiO_2; EtOAc:hexanes 1:4 UV) R_f = 0.15; **m.p.** = 69−74 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.89 (ddd, J = 2.1, 1.6, 0.5 Hz, 1H), 7.78 (ddd, J = 7.8, 1.7, 1.1 Hz, 1H), 7.55 (ddd, J = 8.0, 2.1, 1.1 Hz, 1H), 7.42 (td, J = 7.9, 0.5 Hz, 1H), 6.84 (s, 1H), 3.58 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 190.6, 163.6, 135.4, 134.1, 133.7, 133.5, 130.4, 129.2, 127.5, 113.5, 52.4; **FTIR** (ATR): ν = 1706, 1664, 1597, 1569,1436, 1423, 1330, 1241, 1196, 1176, 1042, 997, 764, 674 cm^{-1}; **HRMS** (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9ClIO_3$ 350.9285; Found 350.9283.

Methyl (E)-4-(3-Bromophenyl)-3-iodo-4-oxobut-2-enoate ***4q***. The title compound was obtained by following the general procedure with 0.24 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 59 mg (62%); **TLC** (SiO_2; EtOAc:hexanes 1:4 UV) R_f = 0.41; **m.p.** = 87−94 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 8.05 (t, J = 1.9 Hz, 1H), 7.82 (ddd, J = 7.8, 1.6, 1.1 Hz, 1H), 7.70 (ddd, J = 8.0, 2.0, 1.1 Hz, 1H), 7.35 (t, J = 7.9 Hz, 1H),

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6.84 (s, 1H), 3.58 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 190.5, 163.6, 137.0, 133.9, 133.5, 132.1, 130.6, 128.0, 123.4, 113.4, 52.4; FTIR (ATR): ν = 1713, 1673, 1603, 1565, 1433, 1421, 1324, 1264, 1232, 1193, 1172, 1066, 1025, 994, 936, 866, 791, 751, 730, 708, 669, 640, 609 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_9BrIO_3$ 394.8780; Found 394.8750.

Methyl (E)-3-Iodo-4-(naphthalen-1-yl)-4-oxobut-2-enoate ***4r.*** The title compound was obtained by following the general procedure with 0.182 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 43.2 mg (64%); TLC (SiO_2; EtOAc:hexanes 5:95 UV) R_f = 0.14; m.p. = 99−104 °C; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 8.36−8.33 (m, 1H), 7.95 (dd, J = 8.6, 1.7 Hz, 1H), 7.91−7.84 (m, 2H), 7.83−7.79 (m, 1H), 7.54 (ddd, J = 8.2, 6.9, 1.3 Hz, 1H), 7.48 (ddd, J = 8.1, 6.9, 1.3 Hz, 1H), 6.85 (s, 1H), 3.47 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 192.0, 163.6, 136.2, 133.2, 132.6, 131.8, 129.9, 129.5, 129.2, 128.0, 127.1, 124.2, 114.6, 52.3; FTIR (ATR): ν = 1712, 1667, 1612, 1434, 1324, 1257, 1222, 1210, 1188, 1164, 1126, 1041, 931, 886, 827, 754, 688, 631, 583 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{12}IO_3$ 366.9831; Found 366.9831.

Methyl (E)-3-Iodo-4-(naphthalen-1-yl)-4-oxobut-2-enoate ***4s.*** The title compound was obtained by following the general procedure with 0.195 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 36.5 mg (51%); TLC (SiO_2; EtOAc:hexanes 5:95 UV) R_f = 0.13; m.p. = 138−153 °C; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 9.13 (d, J = 8.7 Hz, 1H), 8.06 (d, J = 8.2 Hz, 1H), 8.01 (dt, J = 7.2, 0.9 Hz, 1H), 7.89 (dd, J = 8.1, 1.3 Hz, 1H), 7.72−7.66 (m, 1H), 7.60−7.54 (m, 1H), 7.48 (t, J = 7.7 Hz, 1H), 6.85 (s, 1H), 3.47 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 193.2, 163.6, 135.2, 134.2, 132.7, 132.4, 131.4, 129.1, 128.7, 128.6, 127.0, 126.3, 124.3, 116.1, 52.3; FTIR (ATR): ν = 1704, 1651, 1600, 1590, 1435, 1323, 1271, 1234, 1205, 1171, 1100, 1073, 1004, 971, 921, 871, 793, 746, 636 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{12}IO_3$ 366.9831; Found 366.9831.

Methyl (E)-4-(Cyclohex-1-en-1-yl)-3-iodo-4-oxobut-2-enoate ***4t.*** The title compound was obtained by following the general procedure with 0.2 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow liquid; 26.5 mg (41%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.21; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 6.83 (tt, J = 4.0, 1.6 Hz, 1H), 6.67 (s, 1H), 3.64 (s, 3H), 2.32−2.29 (m, 2H), 2.26−2.23 (m, 2H), 1.69−1.59 (m, 4H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 193.3, 163.5, 144.9, 135.5, 132.5, 115.1, 52.2, 26.4, 22.8, 21.7, 21.5; FTIR (ATR): ν = 1711, 1655, 1632, 1604, 1433, 1325, 1268, 1191, 994, 904, 862, 788, 648, 619 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{11}H_{14}IO_3$ 320.9988; Found 321.0020.

Methyl (E)-4-(Cyclohept-1-en-1-yl)-3-iodo-4-oxobut-2-enoate ***4u.*** The title compound was obtained by following the general procedure with 0.21 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Brown liquid; 35 mg (50%); TLC (SiO_2; EtOAc:hexanes 1:4 UV) R_f = 0.41; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 6.97 (t, J = 6.7 Hz, 1H), 6.62 (d, J = 0.9 Hz, 1H), 3.60 (s, 3H), 2.55−2.47 (m, 2H), 2.35−2.27 (m, 2H), 1.74 (p, J = 5.9 Hz, 2H), 1.53−1.44 (m, 4H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 193.3, 163.5, 150.1, 141.7, 132.3, 115.5, 52.2, 32.1, 29.6, 25.8, 25.8, 25.5; FTIR (ATR): ν = 2950, 1716, 1653, 1632, 1600, 1433, 1324, 1264, 1239, 1203, 1166, 1131, 1102, 979, 946, 917, 863, 849 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{12}H_{16}IO_3$ 335.0144; Found 335.0169.

Methyl (E)-3-Bromo-4-oxo-2,4-diphenylbut-2-enoate ***5a.*** The title compound was obtained by following the general procedure with 0.068 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow solid; 17.0 mg (72%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.19; m.p. = 59−64 °C; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 8.03−7.98 (m, 2H), 7.60 (ddt, J = 8.0, 6.8, 1.3 Hz, 1H), 7.51 (ddd, J = 8.5, 7.1, 0.6 Hz, 2H), 7.48−7.42 (m, 3H), 7.39−7.35 (m, 2H), 3.50 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.9, 163.1, 142.3, 138.5, 133.8, 132.8, 129.5, 129.0, 129.0, 128.9, 128.6, 116.8, 52.9; FTIR (ATR): ν = 1712, 1667, 1208, 1173, 1061, 1023, 992, 783, 695, 612 cm^{-1}; HRMS (ESI) m/z: $[M]^+$ calcd for $C_{17}H_{13}BrO_3$ 344.0048; Found 344.0035.

Methyl (E)-4-(3-Bromo-1-methoxy-1,4-dioxo-4-phenylbut-2-en-2-yl)benzoate ***5b.*** The title compound was obtained by following the general procedure with 0.18 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Pale yellow solid; 41.8 mg (58%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.1; m.p. = 64−67 °C; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 8.16 (d, J = 8.7 Hz, 2H), 8.04−7.99 (m, 2H), 7.68−7.62 (m, 1H), 7.57−7.52 (m, 2H), 7.50 (d, J = 8.6 Hz, 2H), 3.95 (s, 3H), 3.55 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 189.8, 166.6, 163.9, 139.6, 135.8, 135.0, 134.2, 133.2, 130.6, 129.8, 129.5, 129.5, 129.1, 53.0, 52.4; FTIR (ATR): ν = 1706, 1664, 1596, 1461, 1449, 1423, 1325, 1265, 1224, 1193, 1181, 1027, 992, 950, 735, 682 cm^{-1}; HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{19}H_{15}BrO_5Na$ 425.0000; Found 425.0009.

Methyl (E)-2-(3,5-Bis(trifluoromethyl)phenyl)-3-bromo-4-oxo-4-phenylbut-2-enoate ***5c.*** The title compound was obtained by following the general procedure with 0.193 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (10:90). White solid; 61 mg (66%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.36; m.p. = 79−85 °C; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 8.03−7.98 (m, 2H), 7.98−7.95 (m, 1H), 7.89 (dd, J = 1.2, 0.6 Hz, 2H), 7.67 (ddt, J = 8.0, 6.9, 1.3 Hz, 1H), 7.60−7.54 (m, 2H), 3.57 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 189.2, 163.1, 137.0, 134.5, 133.9, 132.9, 132.2, 129.9, 129.4, 129.2, 124.2, 122.9, 122.1, 53.2; ^{19}F NMR (471 MHz, $CDCl_3$) δ = −62.67; FTIR (ATR): ν = 1720, 1660, 1539, 1440, 1275, 1173, 1159, 1124, 1099, 1012, 984, 901, 702, 693, 654, 648 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{19}H_{12}BrF_6O_3$ 480.9874; Found 480.9867.

Methyl (E)-2-(3,5-Bis(trifluoromethyl)phenyl)-3-bromo-4-(3-fluorophenyl)-4-oxobut-2-enoate ***5d.*** The title compound was obtained by following the general procedure with 0.12 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (10:90). Pale yellow solid; 31.3 mg (54%); TLC (SiO_2; EtOAc:hexanes 1:10 UV) R_f = 0.38; m.p. = 86−88 °C; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 7.97 (s, 1H), 7.88 (s, 2H), 7.76 (ddd, J = 7.7, 1.6, 1.0 Hz, 1H), 7.70 (ddd, J = 8.9, 2.6, 1.6 Hz, 1H), 7.59−7.52 (m, 1H), 7.37 (tdd, J = 8.2, 2.6, 1.0 Hz, 1H), 3.61 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 187.9, 163.1, 136.7, 136.3, 135.0, 134.2, 131.0, 129.9, 125.2, 123.0, 121.6, 115.9, 53.4; ^{19}F NMR (471 MHz, $CDCl_3$) δ = −62.7; FTIR (ATR): ν = 1705, 1681, 1277, 1254, 1180, 1172, 1094, 1080, 1027, 975, 893, 847, 782, 705, 893, 847, 782, 706, 690, 679, 646 cm^{-1}; HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{19}H_{10}BrF_7O_3Na$ 520.9599; Found 520.9587.

Methyl (E)-3-Iodo-4-oxo-2,4-diphenylbut-2-enoate ***6a.*** The title compound was obtained by following the general procedure with 0.57 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow liquid; 112 mg (50%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.17; ^{1}H NMR (500 MHz, $CDCl_3$) δ = 8.05−8.00 (m, 2H), 7.64−7.59 (m, 1H), 7.53 (ddd, J = 8.1, 6.6, 1.2 Hz, 2H), 7.51−7.46 (m, 2H), 7.46−7.43 (m, 1H), 7.40−7.36 (m, 2H), 3.51 (s, 3H); ^{13}C {^{1}H} NMR (126 MHz, $CDCl_3$) δ = 191.9, 163.1, 142.2, 138.5, 133.9, 132.8, 129.5, 129.0, 129.0, 129.0, 128.6, 116.9, 52.9; FTIR (ATR): ν = 1713, 1671, 1596, 1448, 1433, 1272, 1234, 1211, 1176, 961, 912, 842, 766, 717, 697, 648, 625, 648, 625, 547 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{17}H_{14}IO_3$ 392.9987; Found 392.9992.

Methyl (E)-4-(3-Iodo-1-methoxy-1,4-dioxo-4-phenylbut-2-en-2-yl)benzoate ***6b.*** The title compound was obtained by following the general procedure with 0.1 mmol of the corresponding allenoate. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (5:95). Yellow waxy solid; 22.5 mg (50%); TLC (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.15; ^{1}H NMR (500 MHz,

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$CDCl_3$) δ = 8.08 (d, J = 8.7 Hz, 2H), 7.97−7.91 (m, 2H), 7.62−7.55 (m, 1H), 7.51−7.45 (m, 2H), 7.45−7.41 (m, 2H), 3.89 (s, 3H), 3.48 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 189.8, 166.7, 163.9, 139.6, 135.8, 135.0, 134.2, 133.2, 130.6, 129.8, 129.5, 129.1, 53.0, 52.4; **FTIR** (ATR): ν = 1704, 1667, 1621, 1596, 1428, 1353, 1337, 1273, 1203, 1170, 1123, 1029, 1000, 958, 866, 715, 629, 591 cm^{-1}; **HRMS** (ESI) m/z: [M + Na]$^+$ calcd for $C_{19}H_{15}IO_5Na$ 472.9861; Found 472.9872.

*Synthetic Transformation of **3a/4a**. Synthesis of 4-Oxo-4-phenyl-2-(phenylthio)butanoate **7**.* To an oven-dried Schlenk tube equipped with a Teflon-coated magnetic stir bar was added (*E*)-**3a** (0.2 mmol, 1 equiv), thiophenol (0.2 mmol, 1 equiv), K_2CO_3 (0.1 mmol, 13.8 mg) and MeOH (1 mL, 0.2 M). The reaction was heated to 65 °C using an oil bath mounted on a hot plate magnetic stirrer and stirred for the same temperature until the consumption of the starting material. The reaction mixture was brought to room temperature and concentrated on a rotavapor and the crude product was purified by flash column chromatography using a binary gradient of hexane:ethyl acetate (90:10). The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (10:90). Yellow liquid; 44 mg (73%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.34; **m.p.** = 50−56 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.91 (dd, J = 8.4, 1.3 Hz, 2H), 7.56 (ddt, J = 8.7, 7.0, 1.3 Hz, 1H), 7.53−7.50 (m, 2H), 7.44 (dd, J = 8.1, 7.3 Hz, 2H), 7.35−7.31 (m, 3H), 4.24 (dd, J = 10.1, 4.4 Hz, 1H), 3.69 (s, 3H), 3.67 (d, J = 10.1 Hz, 1H), 3.38 (dd, J = 18.0, 4.4 Hz, 1H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 196.9, 172.2, 136.1, 133.9, 133.7, 132.3, 129.2, 128.8, 128.7, 128.2, 52.6, 45.2, 41.1; **FTIR** (ATR): ν = 1727, 1678, 1580, 1446, 1359, 1333, 1243, 1215, 1165, 1149, 1086, 1024, 997, 982, 957, 758, 745, 737, 713, 686, 649, 615, 557 cm^{-1}; **HRMS** (ESI) m/z: [M + H]$^+$ calcd for $C_{17}H_{17}O_3S$ 301.0898; Found 301.0904.

*Synthesis of Methyl 2,2-Dimethoxy-4-oxo-4-phenylbutanoate **8**.* (*E*)-**4a** (0.26 mmol, 1 equiv) was taken in a headspace vial equipped with a Teflon-coated magnetic stir bar and dissolved in MeOH (1 mL, 0.2 M). To the solution was added Cs_2CO_3 (0.26 mmol, 84.7 mg), and the contents were stirred at room temperature for 5 h. The reaction progress was monitored through TLC to completion. After completion, the reaction mixture was concentrated on a rotavapor. The product was purified by flash column chromatography using a binary gradient of hexane:ethyl acetate (90:10). Title compound **8** was obtained as a white waxy liquid. The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (1:9). 48 mg (73%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.12; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.98−7.90 (m, 2H), 7.62−7.55 (m, 1H), 7.51−7.45 (m, 2H), 3.80 (s, 3H), 3.70 (s, 2H), 3.31 (s, 6H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 195.0, 168.8, 136.5, 133.6, 128.8, 128.2, 100.1, 52.8, 50.2, 42.4; **FTIR** (ATR): ν = 1725, 1710, 1486, 1433, 1390, 1200, 1010, 886, 790, 640 cm^{-1}; **HRMS** (ESI) m/z: [M + H]$^+$ calcd for $C_{13}H_{17}O_5$ 253.1076; Found 253.1084.

*Synthesis of Methyl (Z)-4-Oxo-3,4-diphenylbut-2-enoate **9**.* An oven-dried Schlenk tube equipped with a Teflon-coated magnetic stir bar was evacuated using a vacuum and backfilled with nitrogen three times. (*E*)-**3a** (0.2 mmol, 53.8 mg, 1 equiv) was loaded into the flask and dissolved in Toluene: EtOH (4:1) under the N_2 environment. The stirred contents were loaded with $Pd(PPh_3)_2Cl_2$ (14 mg, 0.02 mmol, 10 mol %), K_2CO_3 (55 mg, 0.4 mmol, 2 equiv), and phenylboronic acid (27 mg, 0.22 mmol, 2 equiv) sequentially. The resulting reaction mixture was stirred at room temperature and the reaction progress was monitored using TLC analysis. The reaction mixture was quenched with H_2O (5 mL) and extracted with EtOAc (5 mL × 3). The combined organic layers were washed with brine, dried over Na_2SO_4, and concentrated on rotavapor. The product was purified by flash column chromatography using a binary gradient of hexane/EtOAc (93/7). The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (1:9). Pale yellow liquid; 41.5 mg (78%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.24; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 7.97−7.92 (m, 2H), 7.55−7.51 (m, 1H), 7.48 (dd, J = 8.1, 1.7 Hz, 2H), 7.42 (dd, J = 8.3, 7.2 Hz, 2H), 7.40−7.32 (m, 3H), 6.51 (s, 1H), 3.61 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 196.6, 165.7, 156.2, 136.0, 134.2, 133.7, 130.7, 129.3, 129.0, 128.9, 127.1, 117.3, 52.0; **FTIR** (ATR): ν = 1712, 1671, 1614, 1596, 1447, 1433, 1350, 1276, 1216, 1193, 1168, 1004, 947, 871, 769, 720, 685, 588, 551 cm^{-1}; **HRMS** (ESI) m/z: [M]$^+$ calcd for $C_{17}H_{15}O_3$ 267.1021; Found 267.1021.

*Synthesis of Methyl (E)-3-((4-Methoxyphenyl)amino)-4-oxo-4-phenylbut-2-enoate **10**.* To a Biotage vial equipped with a Teflon-coated magnetic stir was added a mixture of compound (*E*)-**3a** (26.9 mg, 0.1 mmol, 1 equiv), *p*-anisidine (14.8 mg, 0.12 mmol, 1.2 equiv), $Pd_2(dba)_3$ (2.2 mg, 2.5 mol %), Xantphos (2.9 mg, 5 mol %), and K_2CO_3 (28 mg, 0.2 mmol, 2 equiv) in toluene (1 mL) were stirred at 80 °C using an oil bath mounted on a hot plate magnetic stirrer for 12 h. After completion of the reaction as indicated by TLC, the mixture was cooled to room temperature and purified directly by flash column chromatography on silica gel using a binary gradient of hexane/EtOAc (83/17). The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (17:83). Yellow liquid; 27 mg (86.7%); **TLC** (SiO_2; EtOAc:hexanes 1:9 UV) R_f = 0.19; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 9.98 (s, 1H), 7.94−7.88 (m, 2H), 7.56−7.47 (m, 1H), 7.41−7.35 (m, 2H), 6.82 (d, J = 8.9 Hz, 2H), 6.64 (d, J = 8.9 Hz, 2H), 4.96 (s, 1H), 3.75 (s, 3H), 3.67 (s, 3H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 192.6, 170.5, 156.9, 156.3, 134.9, 134.2, 132.4, 129.7, 128.7, 123.7, 114.5, 89.0, 55.4, 51.1; **FTIR** (ATR): ν = 1658, 1594, 1511, 1448, 1272, 1245, 1205, 1171, 1150, 1034, 972, 943, 909, 826, 789, 727, 688, 640, 615 cm^{-1}; **HRMS** (ESI) m/z: [M + Na]$^+$ calcd for $C_{18}H_{17}NO_4Na$ 334.1055; Found 334.1069.

*Synthesis of Dimethyl (2Z,4Z)-3,4-Dibenzoylhexa-2,4-dienedioate **11**.* An oven-dried Schlenk tube with a magnetic stir bar was sealed with a septum and degassed by vacuum evacuation and nitrogen backfilling. To this tube was added (*E*)-**3a** (53.8 mg, 0.2 mmol, 1 equiv), bis(pinacolato)diboron (25.3 mg, 0.5 equiv), $PdCl_2$dppf.CH_2Cl_2 (9 mol %), KOAc (29 mg, 0.6 mmol, 3 equiv), and dioxane (1 mL) under a N_2 atmosphere. The resulting mixture was stirred at 90 °C using an oil bath mounted on a hot plate magnetic stirrer for 3 h, then filtered through Celite, washed with EtOAc, and evaporated to afford the crude product. The product was purified by flash column chromatography using a binary gradient of hexane/EtOAc (80/20). The desired product was purified by flash column chromatography using eluent EtOAc:hexanes (1:4). White solid; 41 mg (54%); **TLC** (SiO_2; EtOAc:hexanes 1:4 UV) R_f = 0.15; **m.p.** = 173−180 °C; **^{1}H NMR** (500 MHz, $CDCl_3$) δ = 8.02−7.98 (m, 4H), 7.64 (ddt, J = 7.8, 6.9, 1.3 Hz, 2H), 7.55−7.50 (m, 4H), 6.09 (s, 2H), 3.54 (s, 6H); **^{13}C {^{1}H} NMR** (126 MHz, $CDCl_3$) δ = 194.4, 164.6, 150.4, 135.5, 134.4, 129.2, 129.0, 125.0, 52.3; **FTIR** (ATR): ν = 1719, 1667, 1595, 1449, 1435, 1337, 1306, 1235, 1201, 1170, 1038, 1020, 989, 940, 930, 869, 764, 736, 699, 683, 618 cm^{-1}; **HRMS** (ESI) m/z: [M + H]$^+$ calcd for $C_{22}H_{19}O_6$ 379.1181; Found 379.1183.

*Synthesis of 5-Bromo-6-phenylpyridazin-3(2H)-one **12**.* An oven-dried Schlenk tube with a magnetic stir bar was sealed with a septum and degassed by vacuum evacuation and nitrogen backfilling. To this tube was added (*E*)-**3a** (53.8 mg, 0.2 mmol, 1 equiv) and methanol (1 mL). To the solution was added hydrazine hydrate (11 μL, 0.2 mmol, 1 equiv) under nitrogen. The reaction was brought to room temperature and warmed to 60 °C using an oil bath mounted on a hot plate magnetic stirrer. The reaction progress was monitored by TLC analysis to completion. Upon completion, the solvents were removed under rotavapor, and the crude product was purified by flash column chromatography on silica gel using a binary gradient of EtOAc/hexanes (30:70). The desired product was purified by flash column chromatography using eluent EtOAc hexanes (3:7). White solid; 33.5 mg (67%); **TLC** (SiO_2; EtOAc:hexanes 3:7 UV) R_f = 0.12; **m.p.** = 166−175 °C; **^{1}H NMR** (500 MHz, DMSO-d_6) δ = 13.44 (s, 1H), 7.49−7.45 (m, 3H), 7.45−7.41 (m, 3H); **^{13}C {^{1}H} NMR** (126 MHz, DMSO-d_6) δ = 159.7, 146.2, 135.7, 133.3, 130.8, 129.7, 129.7, 128.6; **FTIR** (ATR): ν = 2919, 2849, 1651, 1557, 1494, 1463, 1445, 1424, 1364, 1261, 1181, 1025, 1009, 995, 969, 933, 902, 804, 766, 723, 696, 585 cm^{-1}; **HRMS** (ESI) m/z: [M + H]$^+$ calcd for $C_{10}H_8BrN_2O$ 250.9819; Found 250.9837.

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■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c02951.

> The Supporting Information is available free of charge on the ACS Publications Website. General procedures and characterization data for substrates; procedures for control experiments; NMR spectra for **3a−3u, 4a−4u, 5a−5d, 6a−6b**, and **7−12**; and crystallographic information about **3r**, **5c**, and **11**. Computational details, including the alternative mechanism for electrophilic halogenation, scheme of deprotonation and bromide migration, and the Cartesian coordinates of all intermediates (PDF)

Accession Codes

Deposition Numbers 2359836, 2376974, and 2377429 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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